

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-Q

(Mark One)
☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the quarterly period ended June 30, 2020

Or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission File Number 001-36198

INTERCONTINENTAL EXCHANGE, INC.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

46-2286804

(IRS Employer
Identification Number)

5660 New Northside Drive, 30328

Atlanta, Georgia (Zip Code)

(Address of principal executive offices)

(770) 857-4700

Registrant's telephone number, including area code

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, \$0.01 par value per share	ICE	New York Stock Exchange

Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports); and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of July 27, 2020, the number of shares of the registrant's Common Stock outstanding was 543,276,671 shares.

INTERCONTINENTAL EXCHANGE, INC.

Form 10-Q

Quarterly Period Ended June 30, 2020

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PART I. Financial Statements

Item 1. Consolidated Financial Statements (Unaudited)

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Balance Sheets
(In millions, except per share amounts)
(Unaudited)

	As of June 30, 2020	As of December 31, 2019
Assets:		
Current assets:		
Cash and cash equivalents	\$ 880	\$ 841
Short-term restricted cash and cash equivalents	993	943
Customer accounts receivable, net of allowance for doubtful accounts of \$24 and \$8, respectively	1,279	988
Margin deposits, guaranty funds and delivery contracts receivable	93,960	64,987
Prepaid expenses and other current assets	323	220
Total current assets	97,435	67,979
Property and equipment, net	1,510	1,536
Other non-current assets:		
Goodwill	13,534	13,342
Other intangible assets, net	10,149	10,258
Long-term restricted cash and cash equivalents	408	404
Other non-current assets	1,024	974
Total other non-current assets	25,115	24,978
Total assets	\$ 124,060	\$ 94,493
Liabilities and Equity:		
Current liabilities:		
Accounts payable and accrued liabilities	\$ 520	\$ 505
Section 31 fees payable	316	138
Accrued salaries and benefits	199	291
Deferred revenue	383	129
Short-term debt	705	2,569
Margin deposits, guaranty funds and delivery contracts payable	93,960	64,987
Other current liabilities	110	197
Total current liabilities	96,193	68,816
Non-current liabilities:		
Non-current deferred tax liability, net	2,317	2,314
Long-term debt	7,703	5,250
Accrued employee benefits	190	198
Non-current operating lease liability	274	281
Other non-current liabilities	294	270
Total non-current liabilities	10,778	8,313
Total liabilities	106,971	77,129
Commitments and contingencies		
Redeemable non-controlling interest in consolidated subsidiaries	95	78
Equity:		
Intercontinental Exchange, Inc. stockholders' equity:		

Preferred stock, \$0.01 par value; 100 shares authorized; none issued or outstanding at June 30, 2020 and December 31, 2019	—	—
Common stock, \$0.01 par value; 1,500 shares authorized; 609 and 607 issued at June 30, 2020 and December 31, 2019, respectively, and 544 and 554 shares outstanding at June 30, 2020 and December 31, 2019, respectively	6	6
Treasury stock, at cost; 65 and 53 shares, respectively	(5,050)	(3,879)
Additional paid-in capital	11,856	11,742
Retained earnings	10,462	9,629
Accumulated other comprehensive loss	(321)	(243)
Total Intercontinental Exchange, Inc. stockholders' equity	16,953	17,255
Non-controlling interest in consolidated subsidiaries	41	31
Total equity	16,994	17,286
Total liabilities and equity	<u>\$ 124,060</u>	<u>\$ 94,493</u>

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Income
(In millions, except per share amounts)
(Unaudited)

	Six Months Ended June 30,		Three Months Ended June 30,	
	2020	2019	2020	2019
Revenues:				
Transaction and clearing, net	\$ 2,571	\$ 1,769	\$ 1,207	\$ 907
Data services	1,138	1,099	574	553
Listings	223	222	111	111
Other revenues	149	127	74	63
Total revenues	4,081	3,217	1,966	1,634
Transaction-based expenses:				
Section 31 fees	320	169	154	100
Cash liquidity payments, routing and clearing	807	480	417	236
Total revenues, less transaction-based expenses	2,954	2,568	1,395	1,298
Operating expenses:				
Compensation and benefits	551	507	273	259
Professional services	63	62	34	29
Acquisition-related transaction and integration costs	14	1	2	1
Technology and communication	257	220	126	113
Rent and occupancy	40	35	19	18
Selling, general and administrative	89	83	40	41
Depreciation and amortization	314	315	157	157
Total operating expenses	1,328	1,223	651	618
Operating income	1,626	1,345	744	680
Other income (expense):				
Interest income	8	19	2	10
Interest expense	(156)	(142)	(84)	(71)
Other income, net	31	32	11	9
Other income (expense), net	(117)	(91)	(71)	(52)
Income before income tax expense	1,509	1,254	673	628
Income tax expense	323	284	145	150
Net income	\$ 1,186	\$ 970	\$ 528	\$ 478
Net income attributable to non-controlling interest	(13)	(14)	(5)	(6)
Net income attributable to Intercontinental Exchange, Inc.	\$ 1,173	\$ 956	\$ 523	\$ 472
Earnings per share attributable to Intercontinental Exchange, Inc. common stockholders:				
Basic	\$ 2.14	\$ 1.69	\$ 0.96	\$ 0.84
Diluted	\$ 2.13	\$ 1.68	\$ 0.95	\$ 0.84
Weighted average common shares outstanding:				
Basic	549	565	546	563
Diluted	552	568	549	566

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Comprehensive Income
(In millions)
(Unaudited)

	Six Months Ended June 30,		Three Months Ended June 30,	
	2020	2019	2020	2019
Net income	\$ 1,186	\$ 970	\$ 528	\$ 478
Other comprehensive income (loss):				
Foreign currency translation adjustments, with no tax benefit for both the six months and three months ended June 30, 2020 and 2019	(78)	7	12	(19)
Change in equity method investment	—	(1)	—	—
Other comprehensive income (loss)	(78)	6	12	(19)
Comprehensive income	\$ 1,108	\$ 976	\$ 540	\$ 459
Comprehensive income attributable to non-controlling interest	(13)	(14)	(5)	(6)
Comprehensive income attributable to Intercontinental Exchange, Inc.	\$ 1,095	\$ 962	\$ 535	\$ 453

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)			
	Shares	Value	Shares	Value						
Balance, as of December 31, 2019	607	\$ 6	(53)	\$ (3,879)	\$ 11,742	\$ 9,629	\$ (243)	\$ 31	\$ 17,286	\$ 78
Impact of adoption of ASU 2016-13, net of tax	—	—	—	—	—	(10)	—	—	(10)	—
Other comprehensive loss	—	—	—	—	—	—	(78)	—	(78)	—
Exercise of common stock options	—	—	—	—	22	—	—	—	22	—
Repurchases of common stock	—	—	(12)	(1,099)	—	—	—	—	(1,099)	—
Payments relating to treasury shares	—	—	—	(72)	—	—	—	—	(72)	—
Stock-based compensation	—	—	—	—	73	—	—	—	73	8
Issuance under the employee stock purchase plan	—	—	—	—	16	—	—	—	16	—
Warrants issued to minority interest holders	—	—	—	—	3	—	—	—	3	2
Issuance of restricted stock	2	—	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(15)	(15)	—
Dividends paid to stockholders	—	—	—	—	—	(330)	—	—	(330)	—
Redeemable non-controlling interest	—	—	—	—	—	—	—	—	—	10
Issuance of non-controlling interest	—	—	—	—	—	—	—	9	9	—
Net income (loss) attributable to non-controlling interest	—	—	—	—	—	(13)	—	16	3	(3)
Net income	—	—	—	—	—	1,186	—	—	1,186	—
Balance, as of June 30, 2020	609	\$ 6	(65)	\$ (5,050)	\$ 11,856	\$ 10,462	\$ (321)	\$ 41	\$ 16,994	\$ 95

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)			
	Shares	Value	Shares	Value						
Balance, as of March 31, 2020	609	\$ 6	(61)	\$ (4,647)	\$ 11,810	\$ 10,103	\$ (333)	\$ 26	\$ 16,965	\$ 97
Other comprehensive income	—	—	—	—	—	—	12	—	12	—
Exercise of common stock options	—	—	—	—	11	—	—	—	11	—
Repurchases of common stock	—	—	(4)	(400)	—	—	—	—	(400)	—
Payments relating to treasury shares	—	—	—	(3)	—	—	—	—	(3)	—
Stock-based compensation	—	—	—	—	35	—	—	—	35	(1)
Dividends paid to stockholders	—	—	—	—	—	(164)	—	—	(164)	—
Issuance of non-controlling interest	—	—	—	—	—	—	—	9	9	—
Net income (loss) attributable to non-controlling interest	—	—	—	—	—	(5)	—	6	1	(1)
Net income	—	—	—	—	—	528	—	—	528	—
Balance, as of June 30, 2020	609	\$ 6	(65)	\$ (5,050)	\$ 11,856	\$ 10,462	\$ (321)	\$ 41	\$ 16,994	\$ 95

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest — (Continued)
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)			
	Shares	Value	Shares	Value						
Balance, as of December 31, 2018	604	\$ 6	(35)	\$ (2,354)	\$ 11,547	\$ 8,317	\$ (315)	\$ 30	\$ 17,231	\$ 71
Other comprehensive income	—	—	—	—	—	—	6	—	6	—
Exercise of common stock options	—	—	—	—	17	—	—	—	17	—
Repurchases of common stock	—	—	(10)	(780)	—	—	—	—	(780)	—
Payments relating to treasury shares	—	—	—	(60)	—	—	—	—	(60)	—
Stock-based compensation	—	—	—	—	73	—	—	—	73	2
Issuance under the employee stock purchase plan	—	—	—	—	14	—	—	—	14	—
Issuance of restricted stock	2	—	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(14)	(14)	—
Dividends paid to stockholders	—	—	—	—	—	(312)	—	—	(312)	—
Net income attributable to non-controlling interest	—	—	—	—	—	(14)	—	15	1	(1)
Net income	—	—	—	—	—	970	—	—	970	—
Balance, as of June 30, 2019	606	\$ 6	(45)	\$ (3,194)	\$ 11,651	\$ 8,961	\$ (309)	\$ 31	\$ 17,146	\$ 72

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)			
	Shares	Value	Shares	Value						
Balance, as of March 31, 2019	606	\$ 6	(41)	\$ (2,851)	\$ 11,597	\$ 8,644	\$ (290)	\$ 24	\$ 17,130	\$ 71
Other comprehensive loss	—	—	—	—	—	—	(19)	—	(19)	—
Exercise of common stock options	—	—	—	—	12	—	—	—	12	—
Repurchases of common stock	—	—	(4)	(340)	—	—	—	—	(340)	—
Payments relating to treasury shares	—	—	—	(3)	—	—	—	—	(3)	—
Stock-based compensation	—	—	—	—	40	—	—	—	40	2
Issuance under the employee stock purchase plan	—	—	—	—	2	—	—	—	2	—
Dividends paid to stockholders	—	—	—	—	—	(155)	—	—	(155)	—
Net income attributable to non-controlling interest	—	—	—	—	—	(6)	—	7	1	(1)
Net income	—	—	—	—	—	478	—	—	478	—
Balance, as of June 30, 2019	606	\$ 6	(45)	\$ (3,194)	\$ 11,651	\$ 8,961	\$ (309)	\$ 31	\$ 17,146	\$ 72

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Cash Flows
(In millions)
(Unaudited)

	Six Months Ended June 30,	
	2020	2019
Operating activities:		
Net income	\$ 1,186	\$ 970
Adjustments to reconcile net income to net cash provided by operating activities:		
Depreciation and amortization	314	315
Stock-based compensation	73	64
Deferred taxes	(2)	(42)
Other	(10)	(26)
Changes in assets and liabilities:		
Customer accounts receivable	(305)	(96)
Other current and non-current assets	(39)	3
Section 31 fees payable	178	63
Deferred revenue	253	229
Other current and non-current liabilities	(270)	(98)
Total adjustments	192	412
Net cash provided by operating activities	1,378	1,382
Investing activities:		
Capital expenditures	(48)	(57)
Capitalized software development costs	(96)	(77)
Cash paid for acquisitions, net of cash acquired	(251)	(352)
Return of capital from equity method investment	—	44
Proceeds from investments, net	3	40
Net cash used in investing activities	(392)	(402)
Financing activities:		
Proceeds from debt facilities, net	2,449	—
Repayments of debt facilities	(1,256)	—
Proceeds from/(redemption of) commercial paper, net	(611)	347
Repurchases of common stock	(1,099)	(780)
Dividends to stockholders	(330)	(312)
Payments relating to treasury shares received for restricted stock tax payments and stock option exercises	(72)	(60)
Other	30	23
Net cash used in financing activities	(889)	(782)
Effect of exchange rate changes on cash, cash equivalents and restricted cash and cash equivalents	(4)	1
Net increase (decrease) in cash, cash equivalents, and restricted cash and cash equivalents	93	199
Cash, cash equivalents, and restricted cash and cash equivalents, beginning of period	2,188	1,872
Cash, cash equivalents, and restricted cash and cash equivalents, end of period	\$ 2,281	\$ 2,071
Supplemental cash flow disclosure:		
Cash paid for income taxes	\$ 475	\$ 296
Cash paid for interest	\$ 149	\$ 145

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Notes to Consolidated Financial Statements
(Unaudited)

1. Description of Business

Nature of Business and Organization

We are a leading global operator of regulated exchanges, clearing houses and listings venues, and a provider of data services for commodity, financial, fixed income and equity markets. We operate regulated marketplaces for listing, trading and clearing of a broad array of derivatives contracts and securities across major asset classes. These asset classes include: energy and agricultural commodities, metals, interest rates, equities, exchange-traded funds, or ETFs, credit derivatives, digital assets, bonds and currencies. We also offer mortgage and technology services to the mortgage industry. In addition, we offer comprehensive data services to support the trading, investment, risk management and connectivity needs of customers around the world and across asset classes.

Our exchanges include derivative exchanges in the United States, or U.S., United Kingdom, or U.K., European Union, or EU, Canada and Singapore, and cash equities, equity options and bond trading venues in the U.S. We also operate over-the-counter, or OTC, markets for physical energy, fixed income and credit default swaps, or CDS, trade execution. To serve global derivatives markets, we operate central counterparty clearing houses, or CCPs, in the U.S., U.K., EU, Canada and Singapore (Note 12). We offer a range of data services, globally, for financial and commodity markets, including pricing and reference data, exchange data, analytics, feeds, index services, desktops and connectivity solutions. Through our markets, clearing houses, listings and data services, we provide comprehensive solutions for our customers to manage risk and raise capital through liquid markets, benchmark products, access to capital markets and related services.

2. Summary of Significant Accounting Policies

Basis of Presentation

We prepared the accompanying unaudited consolidated financial statements in accordance with U.S. generally accepted accounting principles, or U.S. GAAP, pursuant to the rules and regulations of the Securities and Exchange Commission, or SEC, regarding interim financial reporting. Accordingly, the unaudited consolidated financial statements do not include all of the information and footnotes required by U.S. GAAP for complete financial statements and should be read in conjunction with our audited consolidated financial statements and related notes thereto for the year ended December 31, 2019. The accompanying unaudited consolidated financial statements reflect all adjustments that are, in our opinion, necessary for a fair presentation of results for the interim periods presented. We believe that these adjustments are of a normal recurring nature.

Preparing financial statements in conformity with U.S. GAAP requires us to make certain estimates and assumptions that affect the amounts that are reported in our consolidated financial statements and accompanying disclosures. Actual amounts could differ from those estimates. The results of operations for the six months and three months ended June 30, 2020 are not necessarily indicative of the results to be expected for any future period or the full fiscal year.

These statements include the accounts of our wholly-owned and controlled subsidiaries. All intercompany balances and transactions between us and our wholly-owned and controlled subsidiaries have been eliminated in consolidation. For consolidated subsidiaries in which our ownership is less than 100% and for which we have control over the assets and liabilities and the management of the entity, the outside stockholders' interests are shown as non-controlling interests. Where outside owners hold an option to require us to repurchase their interests, these amounts are shown as redeemable non-controlling interests and are subject to remeasurement when repurchase is probable.

Recently Adopted Accounting Pronouncements

Standard/Description	Effective Date and Adoption Considerations	Effect on Financial Statements
ASU No. 2016-13, <i>Financial Instruments - Measurement of Credit Losses on Financial Instruments</i> , applies to all financial instruments carried at amortized cost including held-to-maturity debt securities and accounts receivable. It requires financial assets carried at amortized cost to be presented at the net amount expected to be collected and requires entities to record credit losses through an allowance for credit losses on available-for-sale debt securities.	We adopted on January 1, 2020 on a modified retrospective basis.	We evaluated this guidance to determine the impact on our consolidated financial statements. Based on our assessment, we concluded the impact of adoption of this guidance was not material. Further disclosures and details on our adoption are discussed below.
ASU 2017-04, <i>Simplifying the Test for Goodwill Impairment</i> , removes the second step of the goodwill impairment test, which requires a hypothetical purchase price allocation if the fair value of a reporting unit is less than its carrying value. Goodwill impairment will now be measured using the difference between the carrying value and the fair value of the reporting unit, and any loss recognized should not exceed the total amount of goodwill allocated to that reporting unit.	We adopted on January 1, 2020 on a prospective basis.	We evaluated this guidance to determine the impact on our consolidated financial statements. Based on our assessment, we concluded the impact of adoption of this guidance was not material. The fair values of our reporting units have been greater than their corresponding carrying values in recent years. Changes in future projections, market conditions, and other factors may cause a change in the excess of fair value of our reporting units over their corresponding carrying values.
ASU 2018-15, <i>Customer's Accounting for Implementation Costs Incurred in a Cloud Computing Arrangement that is a Service Contract</i> , helps entities evaluate the accounting for fees paid by a customer in a cloud computing arrangement by providing guidance for determining when an arrangement includes a software license and is solely a hosted service. Customers will now apply the same criteria for capitalizing implementation costs as they would for a software license arrangement. The guidance also prescribes the balance sheet, income statement, and cash flow classification of the capitalized implementation costs and related amortization expense, and requires additional quantitative and qualitative disclosures.	We adopted on January 1, 2020 and apply the rules prospectively to eligible costs incurred on or after the effective date.	We evaluated this guidance to determine the impact on our consolidated financial statements. Based on our assessment, we concluded the impact of adoption of this guidance was not material.
ASU No. 2019-12, <i>Simplifying the Accounting for Income Taxes</i> , eliminates certain exceptions related to the approach for intraperiod tax allocation, the methodology for calculating income taxes in an interim period and the recognition of deferred tax liabilities for outside basis differences. It clarifies that single-member limited liability companies, and other similar disregarded entities that are not subject to income tax, are not required to recognize an allocation of consolidated income tax expense in their separate financial statements. Further, it simplifies the accounting for franchise taxes, enacted changes in tax laws or rates and transactions that result in a step-up in the tax basis of goodwill.	Effective for fiscal years beginning after December 15, 2020 with early adoption permitted. We elected early adoption and adopted on January 1, 2020.	We evaluated this guidance to determine the impact on our consolidated financial statements. Based on our assessment, we concluded the impact of adoption of this guidance was not material.

Adoption of ASU 2016-13, Financial Instruments - Measurement of Credit Losses on Financial Instruments

On January 1, 2020, we adopted ASU 2016-13, *Financial Instruments - Measurement of Credit Losses on Financial Instruments*, or ASU 2016-13. This standard requires the application of a current expected credit loss, or CECL, impairment model to financial assets measured at amortized cost, including accounts receivable and certain off-balance-sheet credit exposures. The standard also amends the impairment model for available-for-sale debt securities requiring entities to record credit losses through an allowance account. The CECL model requires an entity to estimate its lifetime expected credit loss and record an allowance that, when deducted from the amortized cost basis of the financial asset, presents the net amount expected to be collected on the financial asset. Adoption of the standard requires more timely recognition of credit losses and credit loss estimates are required to use historical information, current information and reasonable and supportable forecasts of future events.

We adopted ASU 2016-13 using the modified retrospective approach through a cumulative-effect adjustment to retained earnings on January 1, 2020. ASU 2016-13 primarily impacted the calculation of our allowance for doubtful accounts on accounts receivable utilizing the expected credit losses model. Our adoption of ASU 2016-13 was subject to the same internal controls over financial reporting that we apply to our consolidated financial statements and the impact of our adoption was not material. We do not currently hold available-for-sale debt securities, off-balance-sheet credit exposures, or other material financial assets impacted by the standard, besides those mentioned below.

We considered our material financial assets within scope, including our cash equivalents, short-term and long-term restricted cash equivalents as well as our clearing members' cash equivalent and reverse repurchase receivables, and determined that such assets have a de minimis risk of credit loss. We invest our cash and clearing members' cash by placing it in highly-rated government securities, primarily U.S. Treasury securities and other sovereign debt with original maturities of less than three months which we consider to be cash equivalents, or into reverse repurchase agreements, referred to as reverse repos, with primarily overnight maturities. Reverse repos are valued daily and are subject to collateral maintenance provisions whereby the counterparty must provide additional collateral if the value of the underlying securities lose value, in an amount sufficient to maintain collateralization of at least 102%. Therefore, as of and for the six months ended June 30, 2020 we have not recorded a credit loss for these financial assets.

Based on the high turnover and collectability of our accounts receivable, as well as the monthly billing process for the majority of revenue, we did not experience a significant increase in the loss provision recognized upon adoption of the CECL model. Accounts receivable in our futures and clearing businesses have minimal credit risk as all clearing members are pre-screened, collection periods occur within one month and the services to customers are completed almost instantaneously. Our accounts receivable related to market data revenues, cash trading, listing revenues, technology revenues, CDS transaction revenues and bilateral OTC energy transaction revenues subject us to credit losses, but we expeditiously limit our risk of credit loss by taking action such as terminating trading access, terminating public listings or ceasing to distribute data for entities with delinquent accounts. The concentration of risk on our accounts receivable is also mitigated by the high quality and the large number of entities comprising our customer base.

We estimated our allowance for doubtful accounts using an aging method, disaggregated based on major revenue stream categories as well as other unique revenue stream factors. The factors for pooling our accounts receivable balances were specific to each revenue stream based on our risk assessment, past patterns of collectability, our knowledge of the business, and customer-specific situations. We apply estimated reserve percentages to the risk pools identified, which are derived from historical write-off factors that are based on the accounts receivable balance's delinquency status and adjusted as appropriate for our reasonable and supportable estimates of current and future economic conditions. We believe that historical write-off trends provide a basis for estimating future patterns of losses because there have been no significant changes in the mix or risk characteristics of the accounts receivable revenue stream pool populations from the risk pools used to calculate our historical write-off rates. At each measurement date we reassess whether our accounts receivable pools continue to exhibit similar risk characteristics. We then determine if assets need to be isolated further as part of their own specific line item reserve due to specific events, such as a customer's inability to meet its financial obligations (i.e. customer disputes, highly unresponsive customers, delinquency of the receivable, or other indicators of credit deterioration of customers). Lastly, the CECL standard is forward-looking and requires us to factor reasonable and supportable economic expectations into our allowance estimate for the asset's entire expected life, which is generally less than one year.

A reconciliation of the beginning and ending amount of allowance for doubtful accounts is as follows for the six months ended June 30, 2020 (in millions):

	Allowance for Doubtful Accounts
Beginning balance as of December 31, 2019	\$ 8
Impact of adoption of ASU 2016-13	13
Bad debt expense	7
Charge-offs	(4)
Ending balance as of June 30, 2020	\$ 24

The impact of adoption of ASU-2016-13 was \$10 million, net of tax. We recorded this impact as an adjustment to retained earnings on January 1, 2020 as shown in our Consolidated Statement of Changes in Equity and Redeemable Non-Controlling Interest. We have included in our allowance assessment the impact of and our responses to the COVID-19 pandemic. Our bad debt expense in the table above includes that assessment, the impact of which was not material for the six months and three months ended June 30, 2020. We will continue to review our accounts receivable and may incur future charge-offs as better estimates become available in future periods. Charge-offs in the table above represent the write-off of uncollectible receivables, net of recoveries. These amounts also include the impact of foreign currency translation adjustments.

3. Acquisitions

On February 21, 2020, we acquired Bridge2 Solutions, a leading provider of loyalty solutions for merchants and consumers. Bridge2 Solutions enables some of the world's leading brands to engage customers and drive loyalty. It powers incentive and employee perk programs for companies across a wide spectrum of industries. The purchase price has been allocated to the net tangible and identifiable intangible assets and liabilities based on the respective estimated fair values on the date of acquisition. The excess of purchase price over the net tangible and identifiable intangible assets has been recorded as goodwill. Identified intangible assets primarily consist of customer relationships and developed technology, which have been assigned useful lives of twelve years and seven years, respectively. Bridge2 Solutions is included in our Trading and Clearing segment as part of the Bakkt ecosystem.

To fund the acquisition of Bridge2 Solutions, Bakkt completed a capital call for \$300 million in funding by ICE and the minority investors. This acquisition-related capital call triggered a market condition of certain Bakkt equity incentive awards, and as a result, during the six months ended June 30, 2020 we incurred a \$10 million compensation expense related to these awards which has been recorded as an acquisition-related cost.

4. Investments

Our equity investments, including our investments in Euroclear plc, or Euroclear, and Coinbase Global, Inc., among others, are subject to valuation under ASU 2016-01, *Financial Instruments- Overall (Subtopic 825-10): Recognition and Measurement of Financial Assets and Financial Liabilities*. See Note 14 for a discussion of our determination of fair value of our financial instruments.

Investment in OCC

We own a 40% interest in the Options Clearing Corporation, or OCC, which is regulated by the SEC and the Commodity Futures Trading Commission, or CFTC, that we treat as an equity method investment. As of June 30, 2020, the OCC is our only equity method investment and is included in other non-current assets in the accompanying consolidated balance sheet. We recognized \$35 million and \$36 million during the six months ended June 30, 2020 and 2019, respectively, and \$18 million and \$9 million during the three months ended June 30, 2020 and 2019, respectively, of equity earnings as our share of the OCC's estimated profits, which is included in other income. On February 13, 2019, the SEC disapproved the OCC capital plan that had been established in 2015. Following the SEC disapproval, the OCC also announced that it would not be providing a refund to clearing members or declaring a dividend to shareholders for the year ended December 31, 2018, which resulted in higher reported OCC 2018 net income than we had estimated. Therefore, during the six months ended June 30, 2019, we adjusted equity earnings in the OCC by \$19 million in other income to reflect our share of the OCC's 2018 net income. Refer to Note 4 to our consolidated financial statements included in our 2019 Form 10-K for additional details on our OCC investment.

5. Revenue Recognition

Substantially all of our revenues are considered to be revenues from contracts with customers. The related accounts receivable balances are recorded in our balance sheets as customer accounts receivable. We do not have obligations for warranties, returns or refunds to customers, other than rebates, which are settled each period and therefore do not result in variable consideration. We do not have significant revenue recognized from performance obligations that were satisfied in prior periods, and we do not have any transaction price allocated to unsatisfied performance obligations other than in our deferred revenue. Deferred revenue represents our contract liabilities related to our annual, original and other listings revenues as well as certain data services, clearing services and other revenues. See Note 7 for our discussion of deferred revenue balances, activity, and expected timing of recognition. We have elected not to provide disclosures about transaction price allocated to unsatisfied performance obligations if contract durations are less than one year, or if we are not required to estimate the transaction price. For all of our contracts with customers, except for listings and certain data and clearing services, our performance obligations are short term in nature and there is no significant variable consideration. In addition, we have elected the practical expedient of excluding sales taxes from transaction prices. We have assessed the costs incurred to obtain or fulfill a contract with a customer and determined them to be immaterial.

Certain judgments and estimates are used in the identification and timing of satisfaction of performance obligations and the related allocation of transaction price. We believe that these represent a faithful depiction of the transfer of services to our customers. Refer to Note 5 to our consolidated financial statements included in our 2019 Form 10-K where our primary revenue contract classifications are described in detail.

The following table depicts the disaggregation of our revenue according to business line and segment (in millions). Amounts here have been aggregated as they follow consistent revenue recognition patterns, and are consistent with the segment information in Note 15:

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Six Months Ended June 30, 2020:			
Transaction and clearing, net	\$ 2,571	\$ —	\$ 2,571
Data services	—	1,138	1,138
Listings	—	223	223
Other revenues	149	—	149
Total revenues	2,720	1,361	4,081
Transaction-based expenses	1,127	—	1,127
Total revenues, less transaction-based expenses	\$ 1,593	\$ 1,361	\$ 2,954

Timing of Revenue Recognition

Services transferred at a point in time	\$ 1,387	\$ —	\$ 1,387
Services transferred over time	206	1,361	1,567
Total revenues, less transaction-based expenses	\$ 1,593	\$ 1,361	\$ 2,954

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Six Months Ended June 30, 2019:			
Transaction and clearing, net	\$ 1,769	\$ —	\$ 1,769
Data services	—	1,099	1,099
Listings	—	222	222
Other revenues	127	—	127
Total revenues	1,896	1,321	3,217
Transaction-based expenses	649	—	649
Total revenues, less transaction-based expenses	\$ 1,247	\$ 1,321	\$ 2,568

Timing of Revenue Recognition

Services transferred at a point in time	\$ 1,075	\$ —	\$ 1,075
Services transferred over time	172	1,321	1,493
Total revenues, less transaction-based expenses	\$ 1,247	\$ 1,321	\$ 2,568

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Three Months Ended June 30, 2020:			
Transaction and clearing, net	\$ 1,207	\$ —	\$ 1,207
Data services	—	574	574
Listings	—	111	111
Other revenues	74	—	74
Total revenues	1,281	685	1,966
Transaction-based expenses	571	—	571
Total revenues, less transaction-based expenses	\$ 710	\$ 685	\$ 1,395

Timing of Revenue Recognition

Services transferred at a point in time	\$ 621	\$ —	\$ 621
Services transferred over time	89	685	774
Total revenues, less transaction-based expenses	\$ 710	\$ 685	\$ 1,395

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Three Months Ended June 30, 2019:			
Transaction and clearing, net	\$ 907	\$ —	\$ 907
Data services	—	553	553
Listings	—	111	111
Other revenues	63	—	63
Total revenues	970	664	1,634
Transaction-based expenses	336	—	336
Total revenues, less transaction-based expenses	\$ 634	\$ 664	\$ 1,298

Timing of Revenue Recognition

Services transferred at a point in time	\$ 547	\$ —	\$ 547
Services transferred over time	87	664	751
Total revenues, less transaction-based expenses	\$ 634	\$ 664	\$ 1,298

The Trading and Clearing segment revenues above include \$153 million and \$122 million for the six months ended June 30, 2020 and 2019, respectively, and \$64 million and \$62 million for the three months ended June 30, 2020 and 2019, respectively, for services transferred over time related to risk management of open interest performance obligations. A majority of these performance obligations are performed over a short period of time of one month or less.

Beginning in the second quarter of 2019, we have reflected amounts owed under certain third-party revenue share arrangements as technology and communication operating expenses rather than as had been previously recorded net within transaction and clearing revenues. These are included within our Trading and Clearing segment.

6. Goodwill and Other Intangible Assets

The following is a summary of the activity in the goodwill balance for the six months ended June 30, 2020 (in millions):

Goodwill balance at December 31, 2019	\$ 13,342
Acquisitions	219
Foreign currency translation	(31)
Other activity, net	4
Goodwill balance at June 30, 2020	\$ 13,534

The following is a summary of the activity in the other intangible assets balance for the six months ended June 30, 2020 (in millions):

Other intangible assets balance at December 31, 2019	\$	10,258
Acquisitions		67
Foreign currency translation		(35)
Amortization of other intangible assets		(141)
Other intangible assets balance at June 30, 2020	\$	10,149

We completed our acquisition of Bridge2 Solutions during the six months ended June 30, 2020 (Note 3). Foreign currency translation adjustments result from a portion of our goodwill and other intangible assets being held at our U.K., EU and Canadian subsidiaries, whose functional currencies are not the U.S. dollar. The change in other activity, net, in the goodwill table above primarily relates to adjustments to the fair value of the net tangible and intangible assets relating to acquisitions, with a corresponding adjustment to goodwill. We have performed an analysis of impairment indicators, including the economic impact of and our responses to the COVID-19 pandemic, and did not recognize any impairment losses on goodwill or other intangible assets during the six months and three months ended June 30, 2020.

7. Deferred Revenue

Our contract liabilities, or deferred revenue, represent consideration received that is yet to be recognized as revenue. Total deferred revenue was \$465 million as of June 30, 2020, including \$383 million in current deferred revenue and \$82 million in other non-current liabilities. The changes in our deferred revenue during the six months ended June 30, 2020 are as follows (in millions):

	Annual Listings Revenues	Original Listings Revenues	Other Listings Revenues	Data Services and Other Revenues	Total
Deferred revenue balance at December 31, 2019	\$ —	\$ 19	\$ 94	\$ 88	\$ 201
Additions	383	6	25	284	698
Amortization	(191)	(9)	(22)	(212)	(434)
Deferred revenue balance at June 30, 2020	\$ 192	\$ 16	\$ 97	\$ 160	\$ 465

The changes in our deferred revenue during the six months ended June 30, 2019 are as follows (in millions):

	Annual Listings Revenues	Original Listings Revenues	Other Listings Revenues	Data Services and Other Revenues	Total
Deferred revenue balance at December 31, 2018	\$ —	\$ 25	\$ 100	\$ 92	\$ 217
Additions	382	3	24	229	638
Amortization	(191)	(11)	(20)	(186)	(408)
Deferred revenue balance at June 30, 2019	\$ 191	\$ 17	\$ 104	\$ 135	\$ 447

Included in the amortization recognized during the six months ended June 30, 2020 is \$72 million related to the deferred revenue balance as of January 1, 2020. Included in the amortization recognized for the six months ended June 30, 2019 is \$61 million related to the deferred revenue balance as of January 1, 2019. As of June 30, 2020, the remaining deferred revenue balance for original listings revenue, other listings revenue and data services and other revenues will be recognized over the period of time we satisfy our performance obligations as described in Note 5.

8. Debt

Our total debt, including short-term and long-term debt, consisted of the following as of June 30, 2020 and December 31, 2019 (in millions):

	As of June 30, 2020	As of December 31, 2019
Debt:		
Short-term debt:		
Commercial Paper	\$ 700	\$ 1,311
2020 Senior Notes (2.75% senior unsecured notes due December 1, 2020)	—	1,248
Other short-term debt	5	10
Total short-term debt	705	2,569
Long-term debt:		
2022 Senior Notes (2.35% senior unsecured notes due September 15, 2022)	498	497
2023 Senior Notes (3.45% senior unsecured notes due September 21, 2023)	398	398
2023 Senior Notes (4.00% senior unsecured notes due October 15, 2023)	795	794
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,245	1,244
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	496	496
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	592	592
2030 Senior Notes (2.10% senior unsecured notes due June 15, 2030)	1,231	—
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,230	1,229
2050 Senior Notes (3.00% senior unsecured notes due June 15, 2050)	1,218	—
Total long-term debt	7,703	5,250
Total debt	\$ 8,408	\$ 7,819

Credit Facilities

We have a \$3.4 billion senior unsecured revolving credit facility, or the Credit Facility, with a maturity date of August 9, 2023. The Credit Facility includes an option for us to propose an increase in the aggregate amount available for borrowing by up to \$975 million, subject to the consent of the lenders funding the increase and certain other conditions. No amounts were outstanding under the Credit Facility as of June 30, 2020.

As of June 30, 2020, of the \$3.4 billion that is currently available for borrowing under the Credit Facility, \$700 million is required to back-stop the amount outstanding under our U.S. dollar commercial paper program, or the Commercial Paper Program, and \$171 million is required to support certain broker-dealer and other subsidiary commitments. The amount required to back-stop the amounts outstanding under the Commercial Paper Program will fluctuate as we increase or decrease our commercial paper borrowings. The remaining \$2.5 billion is available for working capital and general corporate purposes including, but not limited to, acting as a back-stop to future increases in the amounts outstanding under the Commercial Paper Program.

Our ICE India subsidiary maintains a \$20 million line of credit for its general corporate purposes. As of June 30, 2020, ICE India had borrowed \$5 million, which is reflected as “other short-term debt” in the table above.

Commercial Paper Program

Our Commercial Paper Program is currently backed by the borrowing capacity available under the Credit Facility, as described above. The effective interest rate of commercial paper issuances does not materially differ from short-term interest rates which fluctuate due to market conditions and as a result may impact our interest expense. During the six months ended June 30, 2020, we had net redemptions of \$611 million under the Commercial Paper Program from a portion of the proceeds from the May 26, 2020 issuance of \$2.5 billion in senior notes, discussed below.

Commercial paper notes of \$700 million with original maturities ranging from six to 87 days were outstanding as of June 30, 2020, with a weighted average interest rate of 0.37% per annum, and a weighted average remaining maturity of 42 days.

New Senior Notes

On May 26, 2020, we issued \$2.5 billion in aggregate principal amount of new senior notes. The senior notes comprise \$1.25 billion in aggregate principal amount of 2.10% senior notes due in 2030, or the 2030 Notes, and \$1.25 billion in

aggregate principal amount of 3.00% Senior Notes due in 2050, or the 2050 Notes, and together with the 2030 Notes, the Notes.

We used the net proceeds from the offering of the Notes for general corporate purposes, including to fund the redemption of our \$1.25 billion aggregate principal amount of 2.75% senior notes due in December 2020, or the 2020 Senior Notes, and to pay down a portion of our commercial paper outstanding. Following the pricing of the Notes, we delivered a notice of redemption of the 2020 Senior Notes to Wells Fargo Bank, National Association, as trustee under the indenture governing the 2020 Senior Notes, which was delivered to the holders of the 2020 Senior Notes on May 26, 2020. The 2020 Senior Notes were redeemed on June 25, 2020 in accordance with the terms of the indenture governing the 2020 Senior Notes. In connection with our May 2020 issuance of the Notes and our early redemption of the 2020 Senior Notes, we recorded an extinguishment payment of \$14 million which includes both a make-whole redemption payment and duplicative interest. These costs are included in interest expense in our consolidated statements of income for three months ended June 30, 2020.

We incurred debt issuance costs of \$23 million relating to the issuance of the Notes and these costs are presented in the accompanying consolidated balance sheet as a deduction from the carrying amount of the related debt liability and will be amortized over their respective lives. The Notes contain affirmative and negative covenants, including, but not limited to, certain redemption rights, limitations on liens and indebtedness and limitations on certain mergers, sales, dispositions and lease-back transactions.

9. Share-Based Compensation

We currently sponsor employee and director stock option, restricted stock and employee stock purchase plans. Stock options and restricted stock are granted at the discretion of the Compensation Committee of our Board of Directors, or Board, based on the estimated fair value on the date of grant. The fair value of the stock options and restricted stock on the date of grant is recognized as expense over the vesting period, net of forfeitures. The non-cash compensation expenses recognized in our consolidated statements of income for stock options, restricted stock and under our employee stock purchase plan were \$61 million and \$64 million for the six months ended June 30, 2020 and 2019, respectively, and \$31 million and \$35 million for the three months ended June 30, 2020 and 2019, respectively.

Stock Option Plans

The following is a summary of our stock option activity:

	Number of Options (in thousands)	Weighted Average Exercise Price per Option
Outstanding at December 31, 2019	3,501	\$ 51.87
Granted	413	92.63
Exercised	(490)	45.06
Forfeited	(13)	67.00
Outstanding at June 30, 2020	3,411	57.74

Details of stock options outstanding as of June 30, 2020 were as follows:

	Number of Options (in thousands)	Weighted Average Exercise Price	Weighted Average Remaining Contractual Life (Years)	Aggregate Intrinsic Value (In millions)
Vested or expected to vest	3,411	\$ 57.74	6.2	\$ 116
Exercisable	2,508	\$ 48.98	5.3	\$ 107

Details of stock options exercised are as follows:

<u>Options exercised:</u>	Six Months Ended June 30,		Three Months Ended June 30,	
	2020	2019	2020	2019
Total intrinsic value of options exercised (in millions)	\$ 22	\$ 18	\$ 9	\$ 10

As of June 30, 2020, there were \$11 million in total unrecognized compensation costs related to stock options, which are expected to be recognized over a weighted average period of 1.9 years as the stock options vest.

We use the Black-Scholes option pricing model to value our stock option awards. During the six months ended June 30, 2020 and 2019, we used the assumptions in the table below to compute the value:

Assumptions:	Six Months Ended June 30,	
	2020	2019
Risk-free interest rate	1.46 %	2.49 %
Expected life in years	5.8	5.9
Expected volatility	20 %	20 %
Expected dividend yield	1.30 %	1.44 %
Estimated weighted-average fair value of options granted per share	\$ 16.65	\$ 15.45

The risk-free interest rate is based on the zero-coupon U.S. Treasury yield curve in effect at the date of grant. The expected life is derived from historical and anticipated future exercise patterns. Expected volatility is based on historical volatility data of our stock.

Restricted Stock Plans

Restricted shares are used as an incentive to attract and retain qualified employees and to increase stockholder returns with actual performance linked to both short and long-term stockholder return as well as retention objectives. We issue awards which may contain a combination of time, performance and/or market conditions. The grant date fair value of each award is based on the closing stock price of our stock at the date of grant. For time-based restricted stock, we recognize expense ratably over the vesting period, which is typically three years, net of forfeitures.

In February 2020, we reserved a maximum of 0.9 million restricted shares for potential issuance as performance-based restricted shares to certain of our employees. The number of shares ultimately granted under this award will be based on our actual financial performance as compared to financial performance targets set by our Board and the Compensation Committee for the year ending December 31, 2020, and will also be subject to a market condition reduction based on how our 2020 total stockholder return, or TSR, compares to that of the S&P 500 Index. The maximum compensation expense to be recognized under these performance-based restricted shares is \$82 million if the maximum financial performance target is met and all 0.9 million shares vest. The compensation expense to be recognized under these performance-based restricted shares will be \$41 million if the target financial performance is met, which would result in 0.5 million shares vesting. For these awards with performance conditions, we recognize expense on an accelerated basis over the three-year vesting period based on our quarterly assessment of the probable 2020 actual financial performance as compared to the 2020 financial performance targets. As of June 30, 2020, our best estimate is that the financial performance level will be above target for 2020. Based on this assessment, we recorded non-cash compensation expense of \$13 million and \$8 million for the six months and three months ended June 30, 2020, respectively, related to these awards and the remaining \$41 million in non-cash compensation expense will be recorded on an accelerated basis over the remaining vesting period, including \$16 million which will be recorded over the remainder of 2020.

We also issue awards with a market condition but no performance condition. The fair value of these awards is estimated based on a simulation of various outcomes and includes inputs such as our stock price on the grant date, the valuation of historical awards with market conditions, the relatively low likelihood that the market condition will affect the number of shares granted (as the market condition only affects shares granted in excess of certain financial performance targets), and our expectation of achieving the financial performance targets.

The following is a summary of nonvested restricted shares under all plans discussed above for the six months ended June 30, 2020:

	Number of Restricted Shares (in thousands)	Weighted Average Grant-Date Fair Value per Share
Nonvested at December 31, 2019	3,728	\$ 68.87
Granted	1,351	92.61
Vested	(1,962)	64.79
Forfeited	(58)	74.60
Nonvested at June 30, 2020	3,059	81.86

Performance-based restricted shares have been presented in the table above to reflect the actual shares issued based on the achievement of past performance targets, also considering the impact of any market conditions. Nonvested performance-based restricted shares granted are presented in the table above at the target number of restricted shares that would vest if the performance targets are met.

	Six Months Ended June 30,	
	2020	2019
Time-based restricted stock units granted (in thousands) ⁽¹⁾	778	955
Total fair value of restricted stock vested under all restricted stock plans (in millions)	\$ 178	\$ 160

⁽¹⁾ The remaining shares granted are performance-based.

As of June 30, 2020, there were \$163 million in total unrecognized compensation costs related to time-based and performance-based restricted stock. These costs are expected to be recognized over a weighted-average period of 1.6 years as the restricted stock vests. These unrecognized compensation costs assume that a target performance level will be met on the performance-based restricted shares granted in February 2020.

Bakkt Incentive Units

We sponsor the Bakkt Equity Incentive Plan under which Bakkt issues various Bakkt preferred, common and phantom, or participation, equity unit awards. These awards were made to certain employees and Board members. The units are unvested at the issuance date, are subject to the vesting terms in the award agreements and upon vesting are converted into Bakkt equity or cash. During the three months ended June 30, 2020, we issued additional common and participation unit awards as well as converted certain existing participation unit awards into common unit awards.

During the six months ended June 30, 2020, the \$300 million capital call related to the acquisition of Bridge2 Solutions triggered a market condition of certain of these Bakkt equity incentive awards. The market condition is based on numerous possible Bakkt transaction or event scenarios established on the original date of grant, each of which have a fixed fair market value. Over the life of these awards, we are required to estimate the most likely outcome and reflect the cumulative financial statement impact of any changes between outcomes. As a result, during the six months ended June 30, 2020 we incurred a \$10 million compensation expense related to these awards that has been recorded as an acquisition-related cost.

10. Equity

Stock Repurchase Program

In December 2019, our Board approved an aggregate of \$2.4 billion for future repurchases of our common stock with no fixed expiration date that became effective on January 1, 2020. The \$2.4 billion replaced the previous amount approved by the Board. During the six months ended June 30, 2020, we repurchased 8.8 million shares of our outstanding common stock at a cost of \$800 million under our Rule 10b5-1 trading plan and 3.2 million shares at a cost of \$299 million on the open market during an open trading period. As of June 30, 2020, up to \$1.3 billion capacity remains from the Board authorization for repurchases of our common stock. We expect to fund repurchases from our operating cash flow or borrowings under our debt facilities or our Commercial Paper Program. Repurchases may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. We have entered into a Rule 10b5-1 trading plan, as authorized by our Board, to govern some of the repurchases of our shares of common stock. We may discontinue the stock repurchases at any time and may amend or terminate the Rule 10b5-1 trading plan at any time or enter into additional plans. The approval of our Board for the share repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board may increase or decrease the amount available for repurchases from time to time.

Dividends

During the six months ended June 30, 2020 and 2019, we declared and paid cash dividends per share of \$0.60 and \$0.55, respectively, for an aggregate payout of \$330 million and \$312 million, respectively. During the three months ended June 30, 2020 and 2019, we declared and paid cash dividends per share of \$0.30 and \$0.275, respectively, for an aggregate payout of \$164 million and \$155 million, respectively. The declaration of dividends is subject to the discretion of our Board, and may be affected by various factors, including our future earnings, financial condition, capital requirements, levels of indebtedness, credit ratings, our current and future planned strategic growth initiatives and other considerations that our Board deem relevant. Our Board has adopted a quarterly dividend declaration policy providing that the declaration of any dividends will be determined quarterly by the Board or the Audit Committee, taking into account such factors as our evolving business model, prevailing business conditions and our financial results and capital requirements, without a predetermined annual net income payout ratio.

Accumulated Other Comprehensive Income (Loss)

The following tables present changes in the accumulated balances for each component of other comprehensive income (loss) (in millions):

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2019	\$ (177)	\$ 1	\$ (67)	\$ (243)
Other comprehensive income (loss)	(78)	—	—	(78)
Income tax benefit (expense)	—	—	—	—
Net current period other comprehensive income (loss)	(78)	—	—	(78)
Balance, as of June 30, 2020	\$ (255)	\$ 1	\$ (67)	\$ (321)

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of March 31, 2020	\$ (267)	\$ 1	\$ (67)	\$ (333)
Other comprehensive income (loss)	12	—	—	12
Income tax benefit (expense)	—	—	—	—
Net current period other comprehensive income (loss)	12	—	—	12
Balance, as of June 30, 2020	\$ (255)	\$ 1	\$ (67)	\$ (321)

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2018	\$ (227)	\$ 2	\$ (90)	\$ (315)
Other comprehensive income (loss)	7	(1)	—	6
Income tax benefit (expense)	—	—	—	—
Net current period other comprehensive income (loss)	7	(1)	—	6
Balance, as of June 30, 2019	\$ (220)	\$ 1	\$ (90)	\$ (309)

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of March 31, 2019	\$ (201)	\$ 1	\$ (90)	\$ (290)
Other comprehensive income (loss)	(19)	—	—	(19)
Income tax benefit (expense)	—	—	—	—
Net current period other comprehensive income (loss)	(19)	—	—	(19)
Balance, as of June 30, 2019	\$ (220)	\$ 1	\$ (90)	\$ (309)

11. Income Taxes

Our effective tax rate was 21% and 23% for the six months ended June 30, 2020 and 2019, respectively, and 22% and 24% for the three months ended June 30, 2020 and 2019, respectively. The effective tax rates for the six months and three months ended June 30, 2020 are lower than the effective tax rates for the comparable periods in 2019 primarily due to enacted U.K. tax law that reduced the corporate tax rate from 19% to 17% effective April 1, 2020, as well as state apportionment changes. The U.K. government has subsequently enacted a reversal of this income tax reduction in July 2020 and we will account for the impact of that in the third quarter of 2020.

On March 27, 2020, the Coronavirus Aid, Relief, and Economic Security Act, or CARES Act, was enacted and certain income tax related relief was provided under the CARES Act. There is no material impact of the CARES Act on our income tax provision for the six months and three months ended June 30, 2020.

12. Clearing Operations

We operate six clearing houses, each of which acts as a central counterparty that becomes the buyer to every seller and the seller to every buyer for its clearing members. Through this central counterparty function, the clearing houses provide financial security for each transaction for the duration of the position by limiting counterparty credit risk.

Our clearing houses are responsible for providing clearing services to each of our futures exchanges, and in some cases to third-party execution venues, and are as follows, referred to herein collectively as "the ICE Clearing Houses":

Clearing House	Products Cleared	Exchange where Executed	Location
ICE Clear Europe	Energy, agricultural, interest rates and equity index futures and options contracts and OTC European CDS instruments	ICE Futures Europe, ICE Futures U.S., ICE Endex and third-party venues	U.K.
ICE Clear U.S.	Agricultural, metals, FX and equity index futures and options contracts and digital assets futures contracts	ICE Futures U.S.	U.S.
ICE Clear Credit	OTC North American, European, Asian-Pacific and Emerging Market CDS instruments	Creditex, OTC and third-party venues	U.S.
ICE Clear Netherlands	Derivatives on equities and equity indices traded on regulated markets	ICE Endex	The Netherlands
ICE Clear Singapore	Energy, metals and financial futures products and digital assets futures contracts	ICE Futures Singapore	Singapore
ICE NGX	Physical North American natural gas, electricity and oil futures	ICE NGX	Canada

Original & Variation Margin

Each of the ICE Clearing Houses generally requires all clearing members or participants to deposit collateral in cash or certain pledged assets. The collateral deposits are known as "original margin." In addition, the ICE Clearing Houses may make intraday original margin calls in circumstances where market conditions require additional protection. The daily profits and losses to and from the ICE Clearing Houses due to the marking-to-market of open contracts is known as "variation margin." With the exception of ICE NGX's physical natural gas and physical power products discussed separately below, the ICE Clearing Houses mark all outstanding contracts to market, and therefore pay and collect variation margin, at least once daily.

The amounts that the clearing members and participants are required to maintain are determined by proprietary risk models established by each ICE Clearing House and reviewed by the relevant regulators, independent model validators, risk committees and the boards of directors of the respective ICE Clearing House. The amounts required may fluctuate over time. Each of the ICE Clearing Houses is a separate legal entity and is not subject to the liabilities of the others, or the obligations of the members of the other ICE Clearing Houses.

Should a particular clearing member or participant fail to deposit its original margin or fail to make a variation margin payment, when and as required, the relevant ICE Clearing House may liquidate or hedge its open positions and use their original margin and guaranty fund deposits to pay any amount owed. In the event that the defaulting clearing member's deposits are not sufficient to pay the amount owed in full, the ICE Clearing Houses will first use their respective contributions to the guaranty fund, often referred to as Skin In The Game, or SITG, to pay any remaining amount owed. In the event that the SITG is not sufficient, the ICE Clearing Houses may utilize the respective guaranty fund deposits, or collect limited additional funds from their respective non-defaulting clearing members on a pro-rata basis, to pay any remaining amount owed.

As of June 30, 2020 and December 31, 2019, the ICE Clearing Houses had received or had been pledged \$168.7 billion and \$126.0 billion, respectively, in cash and non-cash collateral in original margin and guaranty fund deposits to cover price movements of underlying contracts for both periods.

Guaranty Funds & ICE Contribution

As described above, mechanisms have been created, called guaranty funds, to provide partial protection in the event of a clearing member default. With the exception of ICE NGX, each of the ICE Clearing Houses requires that each clearing member make deposits into a guaranty fund.

In addition, we have contributed our own capital that could be used if a defaulting clearing member's original margin and guaranty fund deposits are insufficient. Such amounts are recorded as long-term restricted cash and cash equivalents in our balance sheets and are as follows (in millions):

Clearing House	ICE Portion of Guaranty Fund Contribution		Default insurance	
	As of June 30, 2020	As of December 31, 2019	As of June 30, 2020	As of December 31, 2019
ICE Clear Europe	\$237	\$233	\$75	\$75
ICE Clear U.S.	103	103	25	25
ICE Clear Credit	50	50	50	50
ICE Clear Netherlands	2	2	N/A	N/A
ICE Clear Singapore	—	1	N/A	N/A
ICE NGX	15	15	100	100
Total	\$407	\$404	\$250	\$250

Of our total contribution to ICE Clear U.S. above, \$35 million is solely applicable to any losses associated with a default in Bitcoin contracts and other digital asset contracts that ICE Clear U.S. may clear in the future. In April 2020, we increased our contribution to ICE Clear Europe's guaranty fund by \$4 million.

In September 2019, we added a layer of insurance to our clearing member default protection. The default insurance has a three-year term for the following clearing houses in the following amounts: ICE Clear Credit - \$50 million; ICE Clear Europe - \$75 million and, ICE Clear U.S. - \$25 million. The default insurance layer resides after and in addition to the ICE Clear Credit, ICE Clear Europe, and ICE Clear U.S. SITG contributions and before the guaranty fund contributions of the non-defaulting clearing members.

Similar to SITG, the default insurance layer is not intended to replace or reduce the position risk-based amount of the guaranty fund. As a result, the default insurance layer is not a factor that is included in the calculation of the clearing members' guaranty fund contribution requirement. Instead, it serves as an additional, distinct, and separate default resource that should serve to further protect the non-defaulting clearing members' guaranty fund contributions from being mutualized in the event of a default.

As of June 30, 2020, ICE NGX maintains a guaranty fund utilizing a \$100 million letter of credit and a default insurance policy, discussed below.

Cash and Cash Equivalent Deposits

We have recorded cash and cash equivalent margin deposits and amounts due in our balance sheets as current assets with corresponding current liabilities to the clearing members. As of June 30, 2020, our cash and cash equivalent margin deposits are as follows (in millions):

	ICE Clear Europe ⁽¹⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 40,759	\$ 39,147	\$ 5,741	\$ —	\$ 4	\$ 85,651
Unsettled variation margin, net	—	—	—	195	—	195
Guaranty fund	4,414	2,879	516	—	5	7,814
Delivery contracts receivable/payable, net	—	—	—	300	—	300
Total	\$ 45,173	\$ 42,026	\$ 6,257	\$ 495	\$ 9	\$ 93,960

As of December 31, 2019, our cash and cash equivalent margin deposits, are as follows (in millions):

	ICE Clear Europe ⁽²⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 28,318	\$ 22,145	\$ 6,802	\$ —	\$ 2	\$ 57,267
Unsettled variation margin, net	—	—	—	255	—	255
Guaranty fund	4,144	2,268	463	—	5	6,880
Delivery contracts receivable/payable, net	—	—	—	585	—	585
Total	\$ 32,462	\$ 24,413	\$ 7,265	\$ 840	\$ 7	\$ 64,987

⁽¹⁾ \$36.0 billion and \$9.2 billion is related to futures/options and CDS, respectively.

⁽²⁾ \$27.4 billion and \$5.1 billion is related to futures/options and CDS, respectively.

Our cash and cash equivalent margin and guaranty fund deposits are maintained in accounts with national banks and reputable financial institutions or secured through direct investments, primarily in U.S. Treasury securities with original maturities of less than three months, or reverse repurchase agreements with primarily overnight maturities.

To provide a tool to address the liquidity needs of our clearing houses and manage the liquidation of margin and guaranty fund deposits held in the form of cash and high quality sovereign debt, ICE Clear Europe, ICE Clear Credit and ICE Clear U.S. have entered into Committed Repurchase Agreement Facilities, or Committed Repo. Additionally, ICE Clear Credit and ICE Clear Netherlands have entered into Committed FX Facilities to support these liquidity needs. As of June 30, 2020 the following facilities were in place:

- **ICE Clear Europe:** \$1.0 billion in Committed Repo to finance U.S. dollar, euro and pound sterling deposits.
- **ICE Clear Credit:** \$300 million in Committed Repo to finance U.S. dollar and euro deposits, €250 million in Committed Repo to finance euro deposits, and €1.9 billion in Committed FX Facilities to finance euro payment obligations.
- **ICE Clear U.S.:** \$250 million in Committed Repo to finance U.S. dollar deposits.
- **ICE Clear Netherlands:** €10 million in Committed FX Facilities to finance euro payment obligations.

Details of our cash and cash equivalent deposits are as follows (in millions):

Clearing House	Investment Type	As of June 30, 2020	As of December 31, 2019
ICE Clear Europe	National Bank Account ⁽¹⁾	\$ 14,545	\$ 9,667
ICE Clear Europe	Reverse repo	27,262	19,187
ICE Clear Europe	Sovereign Debt	3,362	3,591
ICE Clear Europe	Demand deposits	4	17
ICE Clear Credit	National Bank Account	33,648	19,480
ICE Clear Credit	Reverse repo	4,325	2,411
ICE Clear Credit	Demand deposits	4,053	2,522
ICE Clear U.S.	Reverse repo	4,657	4,320
ICE Clear U.S.	Sovereign Debt	1,600	2,945
Other ICE Clearing Houses	Demand deposits	9	7
ICE NGX	Unsettled Variation Margin and Delivery Contracts Receivable/Payable	495	840
Total		\$ 93,960	\$ 64,987

⁽¹⁾ As of June 30, 2020, ICE Clear Europe held €10.7 billion (\$12.1 billion based on the euro/U.S. dollar exchange rate of 1.1234 as of June 30, 2020) at De Nederlandsche Bank, or DNB, £2.0 billion (\$2.5 billion based on the pound sterling/U.S. dollar exchange rate of 1.2400 as of June 30, 2020) at the Bank of England, or BOE, and €10 million (\$11 million based on the above exchange rate) at the BOE. As of December 31, 2019, ICE Clear Europe held €8.0 billion (\$9.0 billion based on the euro/U.S. dollar exchange rate of 1.1212 as of December 31, 2019) at DNB, £500 million (\$663 million based on the pound sterling/U.S. dollar exchange rate of 1.3260 as of December 31, 2019) at the BOE and €10 million (\$11 million based on the above exchange rate) at the BOE.

Other Deposits

In addition to the cash deposits above, the ICE Clearing Houses have also received other assets from clearing members, which include government obligations, and may include other non-cash collateral such as letters of credit at ICE NGX, or gold on rare occasions at ICE Clear Europe, to mitigate credit risk. For certain deposits, we may impose discount or “haircut” rates to ensure adequate collateral if market values fluctuate. The value-related risks and rewards of these assets remain with the clearing members. Any gain or loss accrues to the clearing member. The ICE Clearing Houses do not, in the ordinary course, rehypothecate or re-pledge these assets. These pledged assets are not reflected in our balance sheets, and are as follows (in millions):

As of June 30, 2020					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 38,546	\$ 11,740	\$ 20,642	\$ —	\$ 70,928
Letters of credit	—	—	—	2,085	2,085
ICE NGX cash deposits	—	—	—	300	300
Total	<u>\$ 38,546</u>	<u>\$ 11,740</u>	<u>\$ 20,642</u>	<u>\$ 2,385</u>	<u>\$ 73,313</u>
Guaranty fund:					
Government securities at face value	<u>\$ 454</u>	<u>\$ 725</u>	<u>\$ 221</u>	<u>\$ —</u>	<u>\$ 1,400</u>
As of December 31, 2019					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 30,635	\$ 13,710	\$ 12,633	\$ —	\$ 56,978
Letters of credit	—	—	—	2,469	2,469
ICE NGX cash deposits	—	—	—	362	362
Total	<u>\$ 30,635</u>	<u>\$ 13,710</u>	<u>\$ 12,633</u>	<u>\$ 2,831</u>	<u>\$ 59,809</u>
Guaranty fund:					
Government securities at face value	<u>\$ 475</u>	<u>\$ 523</u>	<u>\$ 243</u>	<u>\$ —</u>	<u>\$ 1,241</u>

ICE NGX

ICE NGX is the central counterparty to participants on opposite sides of its physically-settled contracts, and the balance related to delivered but unpaid contracts is recorded as a delivery contract net receivable, with an offsetting delivery contract net payable in our balance sheets. Unsettled variation margin equal to the fair value of open contracts is recorded as of each balance sheet date. ICE NGX marks all outstanding contracts to market daily, but only collects variation margin when a clearing member's or participant's open position falls outside a specified percentage of its pledged collateral.

ICE NGX requires participants to maintain cash or letters of credit to serve as collateral in the event of default. The cash is maintained in a segregated bank account, held in trust and remains the property of the participant, therefore, it is not included in our balance sheets. ICE NGX maintains the following accounts with a third-party Canadian chartered bank which are available in the event of physical settlement shortfalls, subject to certain conditions:

Account Type	As of June 30, 2020 (In C\$ millions)	As of June 30, 2020 (In \$USD millions)
Daylight liquidity facility	C\$300	\$221
Overdraft facility	20	15
Total	<u>C\$320</u>	<u>\$236</u>

As of June 30, 2020, ICE NGX maintains a guaranty fund of \$100 million funded by a letter of credit issued by a major Canadian chartered bank, and backed by default insurance underwritten by Export Development Canada, or EDC, a Crown corporation operated at arm's length from the Canadian government. In the event of a participant default where the participant's collateral is depleted, the shortfall would be covered by a draw down on the letter of credit following which ICE NGX would file a claim under the default insurance to recover additional losses up to \$100 million beyond the \$15 million first-loss amount that ICE NGX is responsible for under the default insurance policy.

Clearing House Exposure

Each ICE Clearing House bears financial counterparty credit risk and provides a central counterparty guarantee, or performance guarantee, to its clearing members or participants. To reduce their exposure, the ICE Clearing Houses have a risk management program with both initial and ongoing membership standards. Excluding the effects of original and variation margin, guaranty fund and collateral requirements, the ICE Clearing Houses' maximum estimated exposure for this guarantee is \$150.0 billion as of June 30, 2020, which represents the maximum estimated value by the ICE Clearing Houses of a hypothetical one-day movement in pricing of the underlying unsettled contracts. This value was determined using proprietary risk management software that simulates gains and losses based on historical market prices, volatility and other factors present at that point in time for those particular unsettled contracts. Future actual market price volatility could result in the exposure being significantly different than this amount.

13. Legal Proceedings

We are subject to legal proceedings, claims and investigations that arise in the ordinary course of our business. These include the matters described in Part I, Item 3 "Legal Proceedings" and Note 15 to the consolidated financial statements in Part II, Item 8 of our 2019 Form 10-K, as well as in the comparable sections of our Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (collectively, the "Prior Legal Proceedings Disclosures"). We record estimated expenses and reserves for those matters in circumstances when a loss contingency is considered probable and the related amount is reasonably estimable. Any such accruals may be adjusted as circumstances change. Assessments of losses are inherently subjective and involve unpredictable factors. We do not believe that the resolution of these legal matters, including the matter described below, will have a material adverse effect on our consolidated financial condition, results of operations, or liquidity. It is possible, however, that future results of operations for any particular quarterly or annual period could be materially and adversely affected by any developments relating to the legal proceedings, claims and investigations. A range of possible losses related to the case below cannot be reasonably estimated at this time, except as otherwise disclosed below.

LIBOR Litigation

In the LIBOR litigation, the plaintiffs' appeal of the district court's dismissal of the complaint was docketed by the U.S. Court of Appeals for the Second Circuit. Briefing on the appeal is scheduled to occur between August and December 2020, and the defendants, including ICE and several of its subsidiaries as well as 18 multinational banks and various of their respective subsidiaries and affiliates, will be seeking affirmance of the dismissal. For further information on this matter, please see the Prior Legal Proceedings Disclosures.

14. Fair Value Measurements

Fair value is the price that would be received from selling an asset or paid to transfer a liability (an exit price) in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants at the measurement date. Our financial instruments consist primarily of certain short-term and long-term assets and liabilities, customer accounts receivable, margin deposits and guaranty funds, equity investments, and short-term and long-term debt.

The fair value of our financial instruments is measured based on a three-level hierarchy:

- **Level 1 inputs** — quoted prices for identical assets or liabilities in active markets.
- **Level 2 inputs** — observable inputs other than Level 1 inputs such as quoted prices for similar assets and liabilities in active markets or inputs other than quoted prices that are directly observable.
- **Level 3 inputs** — unobservable inputs supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

Financial assets and liabilities recorded or disclosed at fair value in the accompanying consolidated balance sheets as of June 30, 2020 and December 31, 2019 were classified in their entirety based on the lowest level of input that is significant to the asset or liability's fair value measurement. As of June 30, 2020, we considered the impacts of the COVID-19 pandemic on our fair value assessments, and determined no impairment indicators to be present. We will continue to monitor these fair values in future periods if, or as, circumstances change.

Our mutual funds are equity and fixed income mutual funds held for the purpose of providing future payments for the supplemental executive savings plan and the supplemental executive retirement plan. These mutual funds are classified as equity investments and measured at fair value using Level 1 inputs with adjustments recorded in net income.

MERS is part of our ICE Mortgage Services business and as of December 31, 2019, held fixed income investments as part of a reserve fund in order to satisfy the original terms of the governing documents of our June 2016 acquisition of a majority equity position in MERS. The majority of these investments were held in U.S. Treasuries and measured at fair value using Level 1 inputs with adjustments recorded to other current liabilities. The remaining amount of the reserve fund was held in other fixed income investments and measured using Level 2 inputs. In June 2020, we sold all of the remaining investments and distributed the proceeds to the original MERS shareholders.

Excluding our equity investments without a readily determinable fair value, all other financial instruments are determined to approximate carrying value due to the short period of time to their maturities.

We did not use Level 3 inputs to determine the fair value of assets or liabilities measured at fair value on a recurring basis as of June 30, 2020 or December 31, 2019.

We measure certain assets, such as fixed assets and intangible assets, at fair value on a non-recurring basis. These assets are recognized at fair value if they are deemed to be impaired. During the three months ended June 30, 2020, we recorded certain fixed assets at fair value using Level 2 inputs. As of June 30, 2020 and December 31, 2019, none of our intangible assets were required to be recorded at fair value since no impairments were recorded.

We measure certain equity investments at fair value on a non-recurring basis using our policy election under ASU No. 2016-01, *Financial Instruments-Overall (Subtopic 825-10): Recognition and Measurement of Financial Assets and Financial Liabilities*. During the six months ended June 30, 2020, we evaluated transactions involving these investments and qualitative considerations of impairment as discussed above and concluded that no fair value adjustments were required under this election.

See Note 12 for the fair value considerations related to our margin deposits, guaranty funds and delivery contracts receivable.

The table below displays the fair value of our debt as of June 30, 2020. The fair values of our fixed rate notes were estimated using quoted market prices for these instruments. The fair value of our commercial paper includes a discount and other short-term debt approximates par value since the interest rates on this short-term debt approximate market rates as of June 30, 2020.

	As of June 30, 2020	
	(in millions)	
Debt:	Carrying Amount	Fair value
Commercial Paper	\$ 700	\$ 701
Other short-term debt	5	5
2022 Senior Notes (2.35% senior unsecured notes due September 15, 2022)	498	520
2023 Senior Notes (3.45% senior unsecured notes due September 21, 2023)	398	434
2023 Senior Notes (4.00% senior unsecured notes due October 15, 2023)	795	887
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,245	1,432
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	496	561
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	592	702
2030 Senior Notes (2.10% senior unsecured notes due June 15, 2030)	1,231	1,275
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,230	1,587
2050 Senior Notes (3.00% senior unsecured notes due June 15, 2050)	1,218	1,300
Total debt	\$ 8,408	\$ 9,404

15. Segment Reporting

We operate two reportable business segments: our Trading and Clearing segment and our Data and Listings segment. This presentation is reflective of how our chief operating decision maker reviews and operates our business. Our Trading and Clearing segment comprises our transaction-based execution and clearing businesses. Our Data and Listings segment comprises our data services and our securities listings businesses, which are both largely subscription-based. Our chief operating decision maker does not review total assets or statements of income below operating income by segments; therefore, such information is not presented below. Our two segments do not engage in intersegment transactions.

Financial data for our business segments is as follows for the six months and three months ended June 30, 2020 and 2019 (in millions):

	Six Months Ended June 30, 2020			Six Months Ended June 30, 2019		
	Trading and Clearing Segment	Data and Listings Segment	Consolidated	Trading and Clearing Segment	Data and Listings Segment	Consolidated
Revenues:						
Energy futures and options contracts	\$ 629	\$ —	\$ 629	\$ 484	\$ —	\$ 484
Agricultural and metals futures and options contracts	143	—	143	134	—	134
Financial futures and options contracts	199	—	199	161	—	161
Cash equities and equity options	1,341	—	1,341	800	—	800
Fixed income and credit	233	—	233	167	—	167
OTC and other transactions	26	—	26	23	—	23
Pricing and analytics	—	558	558	—	536	536
Exchange data and feeds	—	363	363	—	356	356
Desktops and connectivity	—	217	217	—	207	207
Listings	—	223	223	—	222	222
Other revenues	149	—	149	127	—	127
Revenues	2,720	1,361	4,081	1,896	1,321	3,217
Transaction-based expenses	1,127	—	1,127	649	—	649
Revenues, less transaction-based expenses	1,593	1,361	2,954	1,247	1,321	2,568
Operating expenses	592	736	1,328	477	746	1,223
Operating income	\$ 1,001	\$ 625	\$ 1,626	\$ 770	\$ 575	\$ 1,345

	Three Months Ended June 30, 2020			Three Months Ended June 30, 2019		
	Trading and Clearing Segment	Data and Listings Segment	Consolidated	Trading and Clearing Segment	Data and Listings Segment	Consolidated
Revenues:						
Energy futures and options contracts	\$ 276	\$ —	\$ 276	\$ 255	\$ —	\$ 255
Agricultural and metals futures and options contracts	59	—	59	72	—	72
Financial futures and options contracts	76	—	76	78	—	78
Cash equities and equity options	672	—	672	410	—	410
Fixed income and credit	111	—	111	80	—	80
OTC and other transactions	13	—	13	12	—	12
Pricing and analytics	—	282	282	—	270	270
Exchange data and feeds	—	183	183	—	180	180
Desktops and connectivity	—	109	109	—	103	103
Listings	—	111	111	—	111	111
Other revenues	74	—	74	63	—	63
Revenues	1,281	685	1,966	970	664	1,634
Transaction-based expenses	571	—	571	336	—	336
Revenues, less transaction-based expenses	710	685	1,395	634	664	1,298
Operating expenses	282	369	651	249	369	618
Operating income	\$ 428	\$ 316	\$ 744	\$ 385	\$ 295	\$ 680

Revenue from two clearing members of the Trading and Clearing segment comprised \$363 million, or 23% of our Trading and Clearing revenues, less transaction-based expenses for the six months ended June 30, 2020 and revenue from one clearing member comprised \$84 million, or 12% for the three months ended June 30, 2020. Revenue from one clearing member of the Trading and Clearing segment comprised \$194 million, or 16% and \$99 million, or 16% of our Trading and Clearing revenues, less transaction-based expenses for the six months and three months ended June 30, 2019, respectively. Clearing members are primarily intermediaries and represent a broad range of principal trading firms. If a clearing member ceased its operations, we believe that the trading firms would continue to conduct transactions and would clear those transactions through another clearing member firm. No additional customers or clearing members accounted for more than 10% of our segment revenues or consolidated revenues for the six months and three months ended June 30, 2020 and 2019.

16. Earnings Per Common Share

The following is a reconciliation of the numerators and denominators of the basic and diluted earnings per common share computations for the six months and three months ended June 30, 2020 and 2019 (in millions, except per share amounts):

	Six Months Ended June 30,		Three Months Ended June 30,	
	2020	2019	2020	2019
Basic:				
Net income attributable to Intercontinental Exchange, Inc.	\$ 1,173	\$ 956	\$ 523	\$ 472
Weighted average common shares outstanding	549	565	546	563
Basic earnings per common share	\$ 2.14	\$ 1.69	\$ 0.96	\$ 0.84
Diluted:				
Weighted average common shares outstanding	549	565	546	563
Effect of dilutive securities - stock options and restricted shares	3	3	3	3
Diluted weighted average common shares outstanding	552	568	549	566
Diluted earnings per common share	\$ 2.13	\$ 1.68	\$ 0.95	\$ 0.84

Basic earnings per common share is calculated using the weighted average common shares outstanding during the period. Common equivalent shares from stock options and restricted stock awards, using the treasury stock method, are included in the diluted per share calculations unless the effect of their inclusion would be antidilutive. During the six months ended June 30, 2020 and 2019, 330,000 and 402,000 outstanding stock options, respectively, were not included in the computation of diluted earnings per common share, because to do so would have had an antidilutive effect. In addition, we have excluded warrants, preferred and common incentive units under the Bakkt Equity Incentive Plan because they are also antidilutive. Certain figures in the table above may not recalculate due to rounding.

17. Subsequent Events

We have evaluated subsequent events and determined that no other events or transactions met the definition of a subsequent event for purposes of recognition or disclosure in the accompanying consolidated financial statements.

ITEM 2. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

In this Quarterly Report on Form 10-Q, unless otherwise indicated, the terms "Intercontinental Exchange," "ICE," "we," "us," "our," "our company" and "our business" refer to Intercontinental Exchange, Inc., together with its consolidated subsidiaries. References to "ICE Products" mean products listed on one or more of our markets. All references to "options" or "options contracts" in the context of our futures products refer to options on futures contracts. Solely for convenience, references in this Quarterly Report to any trademarks, service marks and trade names owned by ICE are listed without the ®, ™ and © symbols, but we will assert, to the fullest extent under applicable law, our rights to these trademarks, service marks and trade names.

We also include references to third-party trademarks, trade names and service marks in this Quarterly Report. Except as otherwise expressly noted, our use or display of any such trademarks, trade names or service marks is not an endorsement or sponsorship and does not indicate any relationship between us and the parties who own such marks and names.

The following discussion should be read in conjunction with our consolidated financial statements and related notes included elsewhere in this Quarterly Report. Due to rounding, figures in tables may not sum exactly.

Forward-Looking Statements

This Quarterly Report on Form 10-Q, including the sections entitled "Notes to Consolidated Financial Statements," "Legal Proceedings" and "Management's Discussion and Analysis of Financial Condition and Results of Operations," contains "forward-looking statements" as defined in the Private Securities Litigation Reform Act of 1995. Any statements contained herein that are not statements of historical fact may be forward-looking statements.

These forward-looking statements relate to future events or our future financial performance and are based on our present beliefs and assumptions as well as the information currently available to us. They involve known and unknown risks, uncertainties and other factors that may cause our results, levels of activity, performance, cash flows, financial position or achievements to differ materially from those expressed or implied by these statements.

Forward-looking statements may be introduced by or contain terminology such as "may," "will," "should," "could," "would," "targets," "goal," "expect," "intend," "plan," "anticipate," "believe," "estimate," "predict," "potential," "continue," or the antonyms of these terms or other comparable terminology. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, cash flows, financial position or achievements. Accordingly, we caution you not to place undue reliance on any forward-looking statements we may make.

Factors that may affect our performance and the accuracy of any forward-looking statements include, but are not limited to, those listed below:

- conditions in global financial markets and domestic and international economic, political and social conditions;
- the impacts of the COVID-19 pandemic on our business, results of operations and financial condition as well as the broader business environment;
- the impact of the introduction of or any changes in laws, regulations, rules or government policies with respect to financial markets, climate change, increased regulatory scrutiny or enforcement actions and our ability to comply with these requirements;
- volatility in commodity prices, equity prices and price volatility of financial benchmarks and instruments such as interest rates, credit spreads, equity indices, foreign exchange rates, and mortgage origination and refinancing trends;
- the business environment in which we operate and trends in our industry, including trading volumes, clearing, data services, fees, changing regulations, competition and consolidation;
- our ability to minimize the risks associated with operating clearing houses in multiple jurisdictions;
- our equity and options exchanges' compliance with their respective regulatory and oversight responsibilities;
- the resilience of our electronic platforms and soundness of our business continuity and disaster recovery plans;
- changes in renewal rates of subscription-based data revenues;
- our ability to execute our growth strategy, identify and effectively pursue, implement and integrate acquisitions and strategic alliances and realize the synergies and benefits of such transactions within the expected time frame;
- the performance and reliability of our trading and clearing technologies and those of third-party service providers;
- our ability to keep pace with technological developments and client preferences;

- our ability to ensure that the technology we utilize is not vulnerable to cyber-attacks, hacking and other cybersecurity risks or other disruptive events or to minimize the impact of any such events;
- our ability to identify trends and adjust our business to benefit from such trends;
- our ability to evolve our benchmarks and indices in a manner that maintains or enhances their reliability and relevance;
- the accuracy of our cost and other financial estimates and our belief that cash flows from operations will be sufficient to service our debt and to fund our operational and capital expenditure needs;
- our ability to incur additional debt;
- our ability to maintain existing market participants and data customers, and to attract new ones;
- our ability to offer additional products and services, leverage our risk management capabilities and enhance our technology in a timely and cost-effective fashion;
- our ability to attract and retain key talent;
- our ability to protect our intellectual property rights and to operate our business without violating the intellectual property rights of others;
- potential adverse results of threatened or pending litigation and regulatory actions and proceedings;
- our ability to realize the expected benefits of our majority investment in Bakkt which could result in additional unanticipated costs and risks; and
- our ability to detect illegal activity such as fraud, money laundering, tax evasion and ransomware scams through digital currency transactions that are easily exploited.

These risks and other factors include those set forth in Part 1, Item 1(A) under the caption “Risk Factors” in our Annual Report on Form 10-K for the year ended December 31, 2019, or our 2019 Form 10-K, as filed with the SEC on February 6, 2020. Due to the uncertain nature of these factors, management cannot assess the impact of each factor on the business or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements.

Any forward-looking statement speaks only as of the date on which such statement is made, and we undertake no obligation to update any of these statements to reflect events or circumstances occurring after the date of this Quarterly Report. New factors may emerge and it is not possible to predict all factors that may affect our business and prospects.

Overview

We are a leading global operator of regulated exchanges, clearing houses and listings venues, and a provider of data services for commodity, financial, fixed income and equity markets. We operate regulated marketplaces for listing, trading and clearing of a broad array of derivatives contracts and securities across major asset classes. These asset classes include: energy and agricultural commodities, metals, interest rates, equities, ETFs, credit derivatives, digital assets, bonds and currencies. We also offer mortgage and technology services to the mortgage industry. In addition, we offer comprehensive data services to support the trading, investment, risk management and connectivity needs of customers around the world and across asset classes.

Our exchanges include derivative exchanges in the U.S., U.K., EU, Canada and Singapore, and cash equities, equity options and bond trading venues in the U.S. We also operate OTC markets for physical energy, fixed income and CDS trade execution. To serve global derivatives markets, we operate central counterparty clearing houses, or CCPs, in the U.S., U.K., EU, Canada and Singapore. We offer a range of data services, globally, for financial and commodity markets, including pricing and reference data, exchange data, analytics, feeds, index services, desktops and connectivity solutions. Through our markets, clearing houses, listings and data services, we provide comprehensive solutions for our customers to manage risk and raise capital through liquid markets, benchmark products, access to capital markets and related services. Our business is conducted as two reportable business segments, our Trading and Clearing segment and our Data and Listings segment, and the majority of our identifiable assets are located in the U.S. and U.K.

Recent Developments

COVID-19

The coronavirus (COVID-19) pandemic has created economic and financial disruptions globally and has led governmental authorities to take unprecedented measures to mitigate the spread of the disease, including travel bans, border closings, business closures, quarantines and shelter-in-place orders, and to take actions designed to stabilize markets and promote economic growth.

From an operational perspective, our businesses, including our exchanges, clearing houses, listing venues, and data services businesses, have remained open and we do not have any plans to close any of our business operations as a

result of the COVID-19 pandemic. However, due to the COVID-19 pandemic, we have taken preventative measures and implemented contingency plans, and currently most of our employees are working remotely. In response to government mandates, we closed all of our office facilities between early March and late April 2020, with only our operationally essential employees working on-site at our facilities for business continuity purposes. As various governments began easing orders requiring office closures in late April, we began a phased re-opening of certain of our office facilities allowing a limited number of operationally non-essential workers to also work on-site. In particular, NYSE, one of our subsidiaries, has partially re-opened its trading floors since initiating fully electronic trading for its exchanges and temporarily closing all trading floors in March 2020. Those employees working on-site are utilizing an alternating framework to allow for social distancing. These measures are in compliance, as necessary, with local government mandates and social distancing directives. We continue to monitor local government mandates in determining our office re-openings, re-closures and work-related travel.

Global health concerns relating to COVID-19 and preventive measures taken to reduce its spread have created significant volatility in the financial markets, which has resulted in higher trading volumes for some of our products and increased demand for our services.

The extent of the impact of the pandemic on our business will depend largely on future developments, including the duration and spread of the outbreak, its severity and the actions taken to contain the disease or treat its impact. We continue to monitor this dynamic situation, including guidance and regulations issued by U.S. and other governmental authorities. In light of the rapidly evolving nature of the COVID-19 outbreak, we are not able at this time to estimate the ultimate effect of the pandemic on our business, results of operations or financial condition in the future.

Acquisition of Bridge2 Solutions

On February 21, 2020, we acquired Bridge2 Solutions, a leading provider of loyalty solutions for merchants and consumers. Bridge2 Solutions enables some of the world's leading brands to engage customers and drive loyalty. It powers incentive and employee perk programs for companies across a wide spectrum of industries.

Regulation

Our activities and the markets in which we operate are subject to regulations that impact us as well as our customers, and, in turn, meaningfully influence our activities, the manner in which we operate and our strategy. We are primarily subject to the jurisdiction of regulatory agencies in the U.S., U.K., EU, Canada, Singapore and Abu Dhabi Global Markets. Failure to satisfy regulatory requirements can or may give rise to sanctions by the applicable regulator.

Global policy makers have undertaken reviews of their existing legal framework governing financial markets in connection with regulatory reform, and have either passed new laws and regulations, or are in the process of debating and/or enacting new laws and regulations that apply to our business and to our customers' businesses. During most of 2020, global policy makers have been focused on the impact of and responses to the COVID-19 pandemic. As a consequence, many regulatory initiatives have been postponed. Legislative and regulatory actions may impact the way in which we or our customers conduct business and may create uncertainty, which could affect trading volumes or demand for market data. See Part 1, Item 1 "Business - Regulation" and Part 1, Item 1(A) "Risk Factors" included in our 2019 Form 10-K for a discussion of the primary regulations applicable to our business and certain risks associated with those regulations.

The key areas in the evolving regulatory landscape that are likely to impact our business are:

- **Brexit timing and implications.** On January 31, 2020, the U.K. officially withdrew from the EU. In connection with the U.K.'s withdrawal from the EU, the U.K. and the EU entered into a withdrawal agreement, which, amongst other things, includes a transitional period until December 31, 2020, during which EU law will continue to apply in and to the U.K.
- **Continued access by EU market participants to U.K. CCPs and exchanges.** Under the terms of the withdrawal agreement, EU law will continue to apply in and to the U.K. for a transitional period until December 31, 2020. During such time, EU market participants will be able to continue clearing through U.K. CCPs such as ICE Clear Europe, and accessing U.K. trading venues, such as ICE Futures Europe. Access by EU market participants to U.K. CCPs following the end of the transitional period will be contingent upon equivalence being granted to the U.K. by the European Commission, or EC, and U.K. CCPs being recognized by the European Securities and Markets Authority, or ESMA. Separately, ICE Futures Europe and ICE Endex will continue to be able to permit access by EU and U.K. persons to transact on their platforms, even in the absence of any trade agreement being entered into by the U.K. and EU prior to the end of the transitional period and/or any trading venue equivalence decisions by the Financial Conduct Authority, or FCA, or the EC. The lack of equivalence decisions for trading venues, however, may result in increased costs for certain EU and U.K. market participants which could impact trading on ICE Futures Europe and ICE Endex. The impact to our business and corresponding regulatory changes remain uncertain at this time. We are monitoring

the impact to our business as a result of these discussions and are pursuing avenues to facilitate continued access for EU and U.K. customers to our services in the event that the transition period ends without any equivalence determination addressing access to CCPs and trading venues being made.

- **The proposed revisions to the regulatory structure of non-EU clearing houses.** On January 1, 2020, the European Market Infrastructure Regulation, or EMIR, 2.2 became effective, which revises the EU's current regulatory and supervisory structure for EU and non-EU clearing houses. These revisions may have an impact on our non-EU clearing houses if they are determined to be systemically important or likely to become systemically important to the financial stability of the EU or one or more of its Member States. In June 2020, the EC published draft delegated regulations on the criteria for determining whether a non-EU clearing house is systemically important to the EU and, for those non-EU clearing houses that are systemically important to the EU, ESMA will assess whether compliance with the clearing house's home country regulation satisfies compliance with EU requirements. The implementation of these delegated regulations could impact one or more of our non-EU clearing houses.
- **Requirement that European exchanges and CCPs offer non-discriminatory access.** The non-discriminatory access provisions of the Markets in Financial Instruments Directive II, or MiFID II, would require our European exchanges and CCPs to offer access to third parties on commercially reasonable terms. In addition, MiFID II could require our European exchanges and CCPs to allow participants to trade and/or clear at other venues, which may encourage competing venues to offer lookalikes of our products. On July 3, 2020, the application of these non-discriminatory access requirements for exchange-traded derivatives was postponed until July 3, 2021.
- **Market data requirements.** Our U.K. and EU derivatives exchanges could be impacted by changes to requirements related to the dissemination of market data. In its December 2019 report to the EC, ESMA recommended against, among other things, outright regulation of market data prices, however ESMA suggested that users could gain transparency into how market data prices are set with the help of new supervisory guidance and targeted changes to the MiFID II/Markets in Financial Instruments Regulation text. The EC is considering ESMA's report and is considering further legislative action in this area.

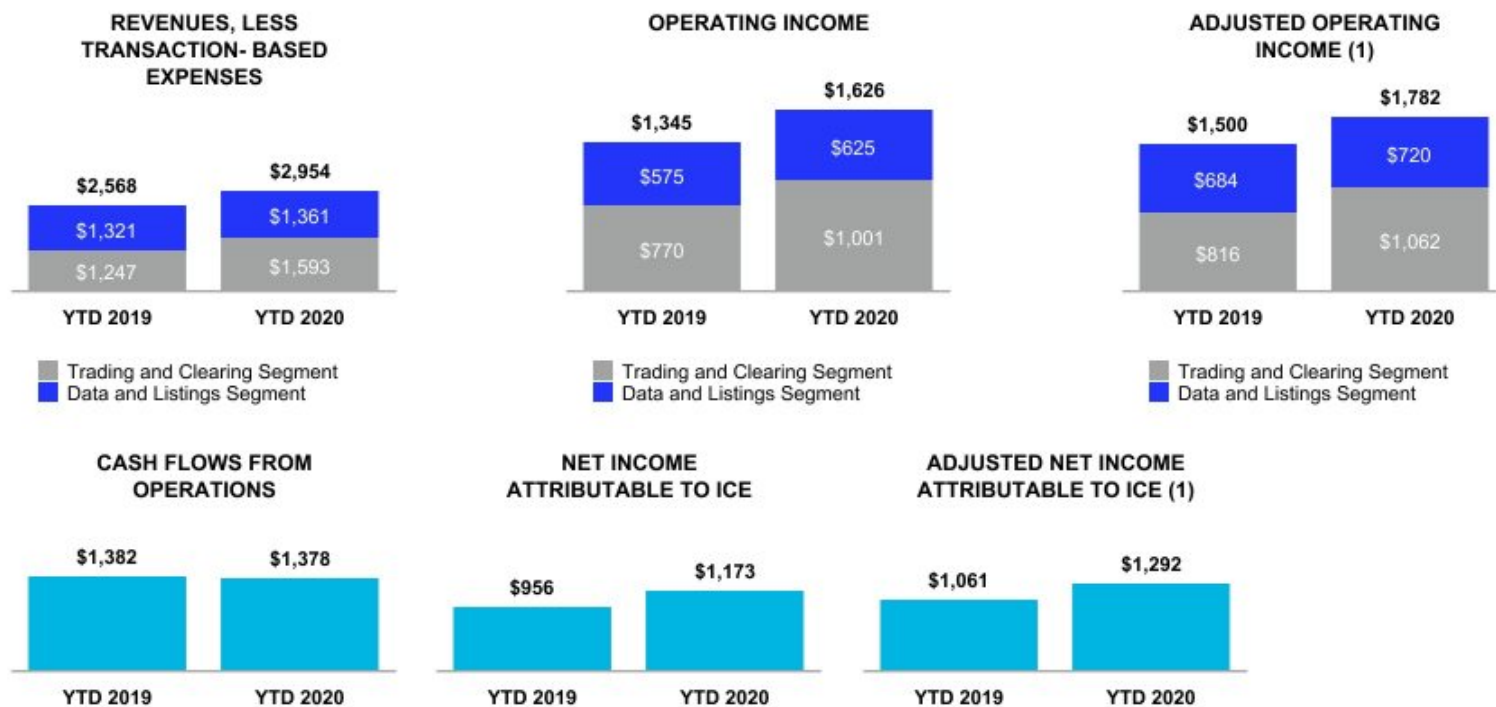
In addition, in October 2019, the SEC proposed a change to Rule 608 that would eliminate the provision allowing market data fee changes proposed by the National Market System Plans, or NMS Plans, to become immediately effective. Further, in May 2020, the SEC finalized an order directing the exchanges and the Financial Industry Markets Authority, or FINRA, to submit to the SEC by August 11, 2020 a proposed new, single NMS Plan to replace the three existing NMS Plans that govern the dissemination of real-time, consolidated equity market data for NMS stocks, or the SIP Order. Once submitted to the SEC, the proposed new, single NMS Plan would not replace the current NMS Plans until it is published for comment and approved by the SEC. Approval of the new SIP Plan by the SEC may affect the NYSE's revenues from the sale of consolidated market data since all the market data fees for consolidated market data would be required to be re-filed with the SEC and would not be effective until approved. In June 2020, the NYSE, Nasdaq and Cboe Global Markets filed petitions for review of the SIP Order with the U.S. Court of Appeals for the District of Columbia Circuit.

- **Position limits.** The adoption and implementation of position limit rules in the U.S. and the EU could have an impact on our commodities business if comparable trading venues in foreign jurisdictions are not subject to equivalent rules. Position limits became effective in the EU beginning January 2018 under MiFID II and are relevant to the markets operated by ICE Futures Europe and ICE Endex. The FCA has published position limits for certain commodity contracts. In April 2020, ESMA issued a report recommending restricting the scope of the position limits regime to critical or significant contracts and the EC continues to work on proposals to revise the position limit framework as part of its ongoing review of MiFID II. In January 2020, the CFTC proposed a rule which replaces the CFTC's prior Dodd-Frank position limit efforts. There is potential for further divergence between MiFID II and U.S. position limit rules if the U.S. makes changes to the financial regulations and the EU either does not make changes to MiFID II or makes changes inconsistent with U.S. regulations.
- **Benchmarks Regulation.** In October 2019, the EC published a consultation reviewing the EU Benchmarks Regulation, or BMR, and included a proposal to provide competent authorities with broader powers to require a benchmark administrator to change the methodology of a critical benchmark, such as LIBOR. In June 2020, the U.K. Government announced its intention to amend the U.K.'s regulatory framework for benchmarks to ensure the FCA has the power to manage and direct any wind-down period prior to a LIBOR cessation, including powers to direct a methodology change for a critical benchmark. Increasing the powers of the FCA or a competent authority to change the methodology of a benchmark or the underlying market represented by a benchmark, such as LIBOR, could result in increased risks to the administrator and users of such benchmark, such as the operator of a derivatives market.
- **The SEC Transaction Fee Pilot.** In December 2018, the SEC adopted a Transaction Fee Pilot. The final rule established a pilot program, for at least one-year and up to two-years, that would limit the fees charged and rebates paid by our five national securities exchanges in certain securities to be designated by the SEC. On June 16, 2020, the U.S. Court of Appeals for the District of Columbia Circuit vacated the Transaction Fee Pilot, finding that the SEC

exceeded its authority. As a result of this ruling, the exchanges will not be required to implement the change to their fees and rebates, as set forth in the Transaction Fee Pilot.

Consolidated Financial Highlights

The following summarizes our results and significant changes in our consolidated financial performance for the periods presented (dollars in millions, except per share amounts and YTD represents the six-month periods ended June 30th):



	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Revenues, less transaction-based expenses	\$ 2,954	\$ 2,568	15 %	\$ 1,395	\$ 1,298	8 %
Operating expenses	\$ 1,328	\$ 1,223	9 %	\$ 651	\$ 618	5 %
Adjusted operating expenses ⁽¹⁾	\$ 1,172	\$ 1,068	10 %	\$ 575	\$ 540	6 %
Operating income	\$ 1,626	\$ 1,345	21 %	\$ 744	\$ 680	10 %
Adjusted operating income ⁽¹⁾	\$ 1,782	\$ 1,500	19%	\$ 820	\$ 758	8 %
Operating margin	55 %	52 %	3 pts	53 %	52 %	1 pt
Adjusted operating margin ⁽¹⁾	60 %	58 %	2 pts	59 %	58 %	1 pt
Other income (expense), net	\$ (117)	\$ (91)	28 %	\$ (71)	\$ (52)	38 %
Income tax expense	\$ 323	\$ 284	14 %	\$ 145	\$ 150	(3) %
Effective tax rate	21 %	23 %	(2 pts)	22 %	24 %	(2 pts)
Net income attributable to ICE	\$ 1,173	\$ 956	23 %	\$ 523	\$ 472	11 %
Adjusted net income attributable to ICE ⁽¹⁾	\$ 1,292	\$ 1,061	22 %	\$ 584	\$ 534	9 %
Diluted earnings per share attributable to ICE common stockholders	\$ 2.13	\$ 1.68	27 %	\$ 0.95	\$ 0.84	13 %
Adjusted diluted earnings per share attributable to ICE common stockholders ⁽¹⁾	\$ 2.34	\$ 1.87	25 %	\$ 1.07	\$ 0.94	14 %
Cash flows from operating activities	\$ 1,378	\$ 1,382	— %			

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. Adjusted net income attributable to ICE and adjusted diluted earnings per share attributable to ICE common stockholders are presented net of taxes. These adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

- Revenues, less transaction-based expenses, increased \$386 million and \$97 million for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019. See "-Trading and Clearing Segment" and "Data and Listings Segment" below for a discussion of the significant changes in our revenues. The increase in revenues during the six months and three months ended June 30, 2020 includes \$10 million and \$5 million, respectively, in unfavorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2019. See Item 3 "Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk" below for additional information on the impact of currency fluctuations.
- Operating expenses increased \$105 million and \$33 million for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019. See "-Consolidated Operating Expenses" below for a discussion of the significant changes in our operating expenses. The increase in operating expenses during the six months and three months ended June 30, 2020 includes \$4 million and \$2 million, respectively, in favorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2019. See Item 3 "Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk" below for additional information on the impact of currency fluctuations.

Variability in Quarterly Comparisons

Our business environment has been characterized by:

- globalization of marketplaces, customers and competitors;
- commodity, interest rate and financial markets uncertainty;
- growing demand for data to inform customers' risk management and investment decisions;
- evolving, increasing and disparate regulation across multiple jurisdictions;
- price volatility increasing customers' demand for risk management services;
- increasing focus on capital and cost efficiencies;
- customers' preference to manage risk in markets demonstrating the greatest depth of liquidity and product diversity;
- the evolution of existing products and new product innovation to serve emerging customer needs and changing industry agreements;
- rising demand for speed, data, data capacity and connectivity by market participants, necessitating increased investment in technology; and
- consolidation and increasing competition among global markets for trading, clearing and listings.

For additional information regarding the factors that affect our results of operations, see Item 1(A) "Risk Factors" included in our 2019 Form 10-K, and Part II, Item 1(A) "Risk Factors" below.

Segment Results

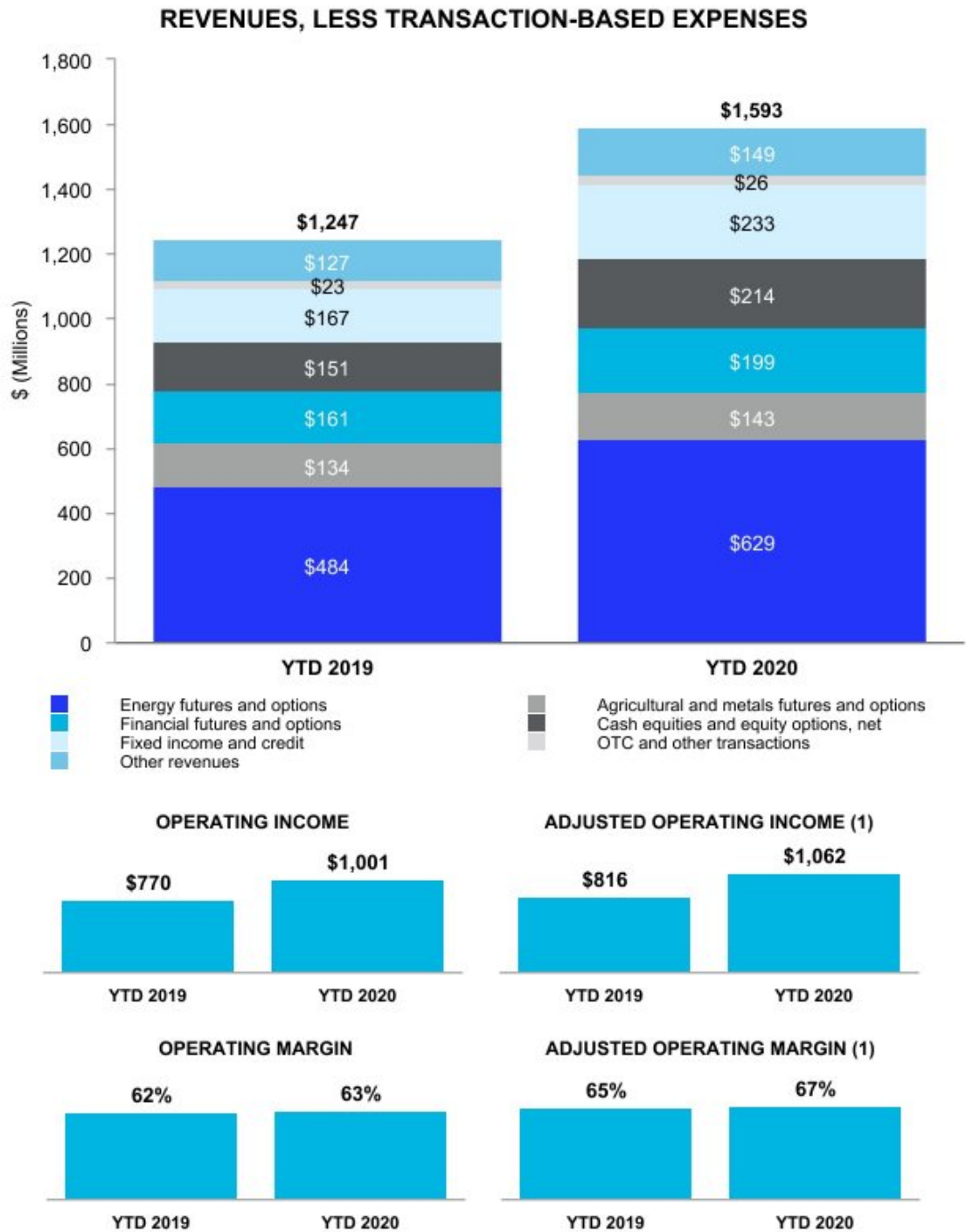
Our business is conducted through two reportable business segments:

- Trading and Clearing, which comprises our transaction-based execution and clearing businesses; and
- Data and Listings, which comprises our subscription-based data services and securities listings businesses.

While revenues are recorded specifically in the segment in which they are earned or to which they relate, a significant portion of our operating expenses are not solely related to a specific segment because the expenses serve functions that are necessary for the operation of both segments. We use a pro-rata revenue approach as the allocation method for the expenses that do not relate solely to one segment. Further, we did not allocate expenses to specific revenue streams within these segments since such an allocation is not reasonably possible. Our two segments do not engage in intersegment transactions.

Trading and Clearing Segment

The following presents selected statements of income data for our Trading and Clearing segment (dollars in millions and YTD represents the six-month periods ended June 30th):



(1) The adjusted numbers in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Revenues:						
Energy futures and options contracts	\$ 629	\$ 484	30 %	\$ 276	\$ 255	8 %
Agricultural and metals futures and options contracts	143	134	7	59	72	(18)
Financial futures and options contracts	199	161	23	76	78	(3)
Cash equities and equity options	1,341	800	68	672	410	64
Fixed income and credit	233	167	40	111	80	38
OTC and other transactions	26	23	13	13	12	14
Transaction and clearing, net	2,571	1,769	45	1,207	907	33
Other revenues	149	127	17	74	63	17
Revenues	2,720	1,896	43	1,281	970	32
Transaction-based expenses	1,127	649	74	571	336	70
Revenues, less transaction-based expenses	1,593	1,247	28	710	634	12
Other operating expenses	448	359	25	216	189	14
Depreciation and amortization	130	117	12	64	59	10
Acquisition-related transaction and integration costs	14	1	n/a	2	1	57
Operating expenses	592	477	24	282	249	13
Operating income	\$ 1,001	\$ 770	30 %	\$ 428	\$ 385	11 %

Transaction and Clearing Revenues

Our transaction and clearing revenues consist of fees collected from our derivatives, fixed income, cash equities and equity options trading, derivatives clearing and mortgage technology services. Because transaction and clearing revenues are generally assessed on a per-contract basis, revenues and profitability fluctuate with changes in contract volume and due to product mix. Rates per-contract, or RPC, are driven by the number of contracts or securities traded and the fees charged per contract, net of certain rebates. Our per-contract transaction and clearing revenues will depend upon many factors, including, but not limited to, market conditions, transaction and clearing volume, product mix, pricing, applicable revenue sharing and market making agreements, and new product introductions.

For the six months ended June 30, 2020 and 2019, 19% and 20%, respectively, of our Trading and Clearing segment revenues, less transaction-based expenses, were billed in pounds sterling or euros and for the three months ended June 30, 2020 and 2019, 17% and 19%, respectively, of our Trading and Clearing segment revenues, less transaction-based expenses, were billed in pounds sterling or euros. For the six months and three months ended June 30, 2020 as compared to the same periods in 2019, foreign currency fluctuations due to the strengthening of the U.S. dollar compared to the pound sterling and euro resulted in a decrease of \$8 million and \$4 million, respectively, to our Trading and Clearing segment revenues, less transaction-based expenses. See Item 3 “- Quantitative and Qualitative Disclosures About Market Risk -Foreign Currency Exchange Rate Risk” below for additional information on the impact of currency fluctuations.

Our transaction and clearing revenues are presented net of rebates. We offer rebates in certain of our markets primarily to support market liquidity and trading volume by providing qualified participants in those markets a discount to the applicable commission rate. Such rebates are calculated based on volumes traded. We recorded rebates of \$525 million and \$435 million for the six months ended June 30, 2020 and 2019, respectively, and \$219 million and \$220 million for the three months ended June 30, 2020 and 2019, respectively. The increase in rebates for the six months ended June 30, 2020 from the comparable period in 2019 was primarily due to the increase in volumes traded. Rebates were flat for the three months ended June 30, 2020 from the comparable period in 2019.

- **Energy Futures and Options Contracts:** Total energy volume increased 34% and revenues increased 30% for the six months ended June 30, 2020 from the comparable period in 2019 and volume increased 14% and revenues increased 8% for the three months ended June 30, 2020 from the comparable period in 2019.
 - Total oil volume increased 33% for the six months ended June 30, 2020 from the comparable period in 2019 and increased 5% for the three months ended June 30, 2020 from the comparable period in 2019 due to increased risk management activity driven by shifting supply/demand dynamics related to various geopolitical events and the emergence of COVID-19.

- Our global natural gas futures and options volume increased 41% and 40% for the six months and three months ended June 30, 2020, respectively. The volume increase in our North American natural gas products was primarily driven by shifting supply/demand dynamics and various geopolitical events. The strength in our European TTF gas volumes was driven by the continued emergence of TTF as not only the European benchmark, but also the emerging global benchmark, for natural gas as the commodity continues to globalize.
- **Agricultural and Metals Futures and Options Contracts:** In our agricultural and metals futures and options markets total volume increased 6% for the six months ended June 30, 2020 and decreased 18% for the three months ended June 30, 2020 from the comparable periods in 2019 and revenues increased 7% for the six months ended June 30, 2020 and decreased 18% for the three months ended June 30, 2020 from the comparable periods in 2019. The overall increase in agricultural volumes for the six months ended June 30, 2020 was primarily driven by price volatility resulting from shifting supply/demand dynamics related to COVID-19, the sharp decline in global oil prices and weather concerns. The overall decrease in agricultural volumes for the three months ended June 30, 2020 was primarily driven by lower commodity price volatility than the prior year period.
 - Sugar futures and options volumes increased 19% and decreased 6% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019.
 - Other agricultural and metal futures and options volume decreased 5% and 27% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019.
- **Financial Futures and Options Contracts:** In our financial futures and options markets total volume increased 12% and decreased 13%, respectively, for the six months and three months ended June 30, 2020 from the comparable periods in 2019 and revenues increased 23% and decreased 3%, respectively, for the six months and three months ended June 30, 2020 from the comparable periods in 2019.
 - Interest rate futures and options volume increased 8% and decreased 17% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019 and revenue increased 13% and decreased 14% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019. The interest rate futures and options volume increase for the six months ended June 30, 2020 was driven, in part, by uncertainty related to the economic impact of COVID-19. The interest rate futures and options volume decrease for the three months ended June 30, 2020 was primarily driven by the impact of quantitative easing measures implemented by major central banks in response to COVID-19. Interest rate futures and options revenues were \$111 million and \$97 million for the six months ended June 30, 2020 and 2019, respectively, and \$39 million and \$44 million for the three months ended June 30, 2020 and 2019, respectively.
 - Other financial futures and options volume, which includes our MSCI®, FTSE® and NYSE FANG+ equity index products, increased 29% and 8% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019. Other financial futures and options volume increased for the six months and three months ended June 30, 2020, due to increased equity market volatility driven by uncertainty related to COVID-19. Other financial futures and options revenues were \$88 million and \$64 million for the six months ended June 30, 2020 and 2019, respectively, and \$37 million and \$34 million for the three months ended June 30, 2020 and 2019, respectively.
- **Cash Equities and Equity Options:** Cash equities handled volume increased 53% and 61% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019 due to heightened volatility driven by uncertainty related to COVID-19 and various geopolitical events. Cash equities revenues, net of transaction-based expenses, were \$163 million and \$102 million for the six months ended June 30, 2020 and 2019, respectively, and \$80 million and \$50 million for the three months ended June 30, 2020 and 2019, respectively.

Equity options volume increased 42% and 44% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019 primarily due to higher industry volumes and heightened volatility driven by uncertainty related to COVID-19 and various geopolitical events. Equity options revenues, net of transaction-based expenses, were \$51 million and \$49 million for the six months ended June 30, 2020 and 2019, respectively, and \$21 million and \$24 million for the three months ended June 30, 2020 and 2019, respectively.
- **Fixed Income and Credit:** Fixed income and credit includes revenues from ICE Mortgage Services, ICE Bonds and CDS execution and clearing.

CDS clearing revenues were \$84 million and \$67 million for the six months ended June 30, 2020 and 2019, respectively, and \$32 million and \$29 million for the three months ended June 30, 2020 and 2019, respectively. The notional value of CDS cleared was \$11.0 trillion and \$7.6 trillion for the six months ended June 30, 2020 and 2019, respectively, and \$3.9 trillion and \$3.2 trillion for the three months ended June 30, 2020 and 2019, respectively. The increase in CDS clearing revenues for the six months and three months ended June 30, 2020 from the comparable periods in 2019 included record buy-side activity in terms of index notional amount cleared during the period driven by heightened market volatility related to COVID-19.

Mortgage services revenues were \$98 million and \$48 million for the six months ended June 30, 2020 and 2019, respectively, and \$54 million and \$27 million for the three months ended June 30, 2020 and 2019, respectively. The increase in mortgage services revenues was primarily driven by the acquisition of Simplifile in June 2019.

- **OTC and Other Transactions:** OTC and other transactions include revenues from our OTC energy business and other trade confirmation services.

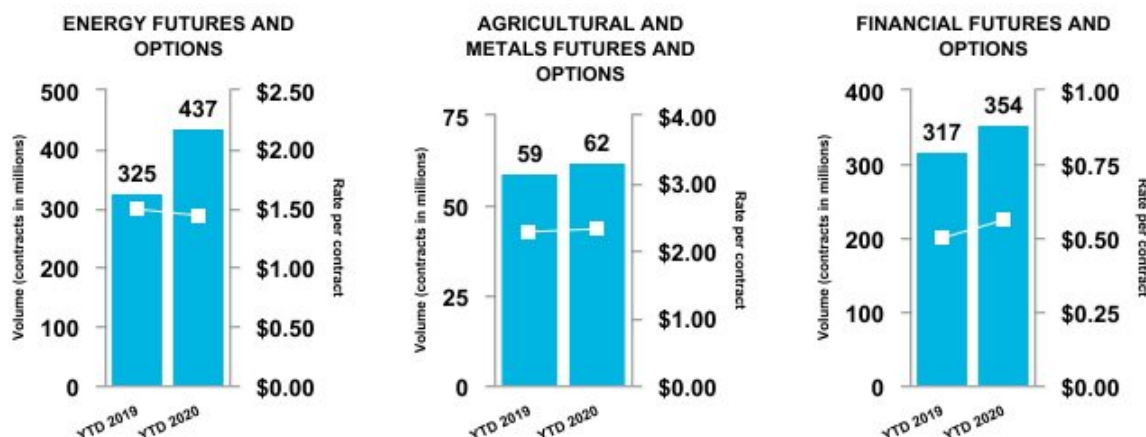
Other Revenues

Other revenues primarily include interest income on certain clearing margin deposits, regulatory penalties and fines, fees for use of our facilities, regulatory fees charged to member organizations of our U.S. securities exchanges, designated market maker service fees, technology development fees, exchange membership fees and agricultural grading and certification fees. The increase in other revenues for the six months and three months ended June 30, 2020 from the comparable periods in 2019 was primarily due to increased interest income earned on certain clearing margin deposits reflecting higher balances, increased regulatory fees and the acquisition of Bridge2 Solutions.

Selected Operating Data

The following charts and tables present trading activity in our futures and options markets by commodity type based on the total number of contracts traded, as well as futures and options rate per contract (in millions, except for percentages and rate per contract amounts and YTD represents the six-month periods ended June 30th):

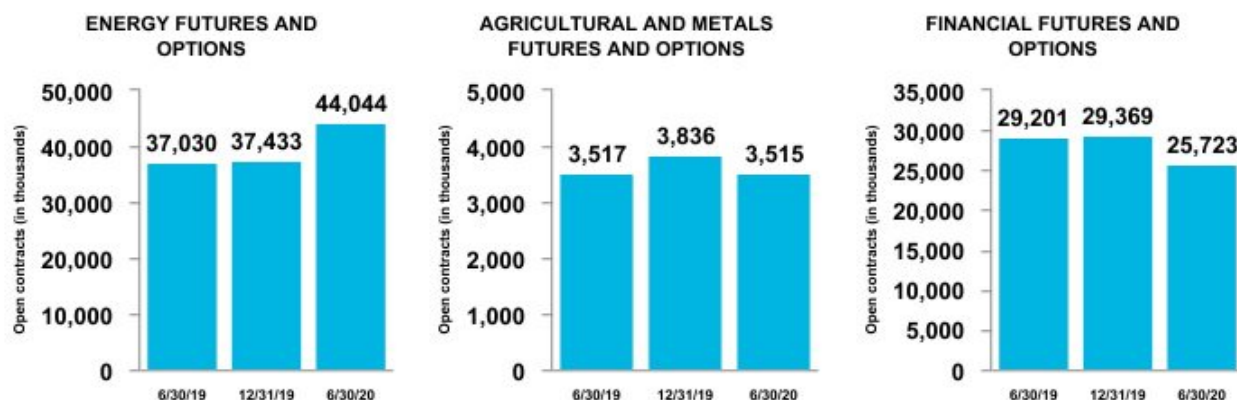
Volume and Rate per Contract



	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Number of contracts traded (in millions):						
Energy futures and options	437	325	34 %	193	169	14 %
Agricultural and metals futures and options	62	59	6	25	32	(18)
Financial futures and options	354	317	12	131	149	(13)
Total	853	701	22 %	349	350	— %
	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Average Daily Volume of contracts traded (in thousands):						
Energy futures and options	3,499	2,624	33 %	3,068	2,694	14 %
Agricultural and metals futures and options	497	474	5	408	498	(18)
Financial futures and options	2,795	2,521	11	2,069	2,369	(13)
Total	6,791	5,619	21 %	5,545	5,561	— %
	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Rate per contract:						
Energy futures and options	\$ 1.44	\$ 1.49	(3) %	\$ 1.42	\$ 1.50	(5) %
Agricultural and metals futures and options	\$ 2.31	\$ 2.28	1 %	\$ 2.33	\$ 2.31	1 %
Financial futures and options	\$ 0.56	\$ 0.50	11 %	\$ 0.58	\$ 0.51	12 %

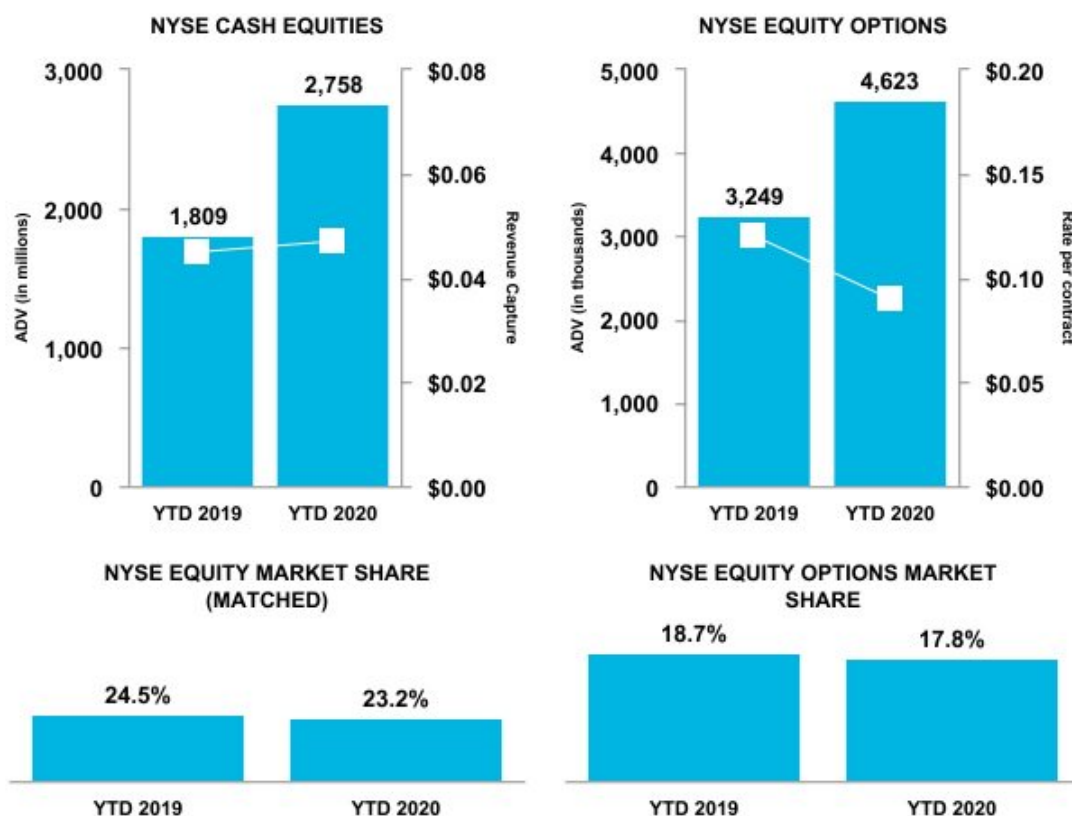
Open interest is the aggregate number of contracts (long or short) that clearing members hold either for their own account or on behalf of their clients. Open interest refers to the total number of contracts that are currently “open,” – in other words, contracts that have been traded but not yet liquidated by either an offsetting trade, exercise, expiration or assignment. Open interest is also a measure of the future activity remaining to be closed out in terms of the number of contracts that members and their clients continue to hold in the particular contract and by the number of contracts held for each contract month listed by the exchange. The following charts and table present our quarter-end open interest for our futures and options contracts (in thousands, except for percentages):

Open Interest



	As of June 30,		
	2020	2019	Change
Open interest — in thousands of contracts:			
Energy futures and options	44,044	37,030	19 %
Agricultural and metals futures and options	3,515	3,517	—
Financial futures and options	25,723	29,201	(12)
Total	73,282	69,748	5 %

The following charts and table present selected cash and equity options trading data (all trading volume below is presented as average net daily trading volume, or ADV, and is single counted and YTD represents the six-month periods ended June 30th):



	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
NYSE cash equities:						
Total cash handled volume (shares in millions)	2,758	1,809	53 %	2,782	1,733	61 %
Total cash market share matched	23.2 %	24.5 %	(1.3) pts	22.1 %	24.4 %	(2.3) pts
NYSE equity options (contracts in thousands):						
NYSE equity options volume	4,623	3,249	42 %	4,566	3,169	44 %
Total equity options volume	25,992	17,329	50 %	26,643	17,327	54 %
NYSE share of total equity options	17.8 %	18.7 %	(1.0) pts	17.1 %	18.3 %	(1.2) pts
Revenue capture or rate per contract:						
Cash equities rate per contract (per 100 shares)	\$0.047	\$0.045	4 %	\$0.046	\$0.046	— %
Equity options rate per contract	\$0.09	\$0.12	(27) %	\$0.07	\$0.12	(39) %

Handled volume represents the total number of shares of equity securities, ETFs and crossing session activity internally matched on our exchanges or routed to and executed on an external market center. Matched volume represents the total number of shares of equity securities, ETFs and crossing session activity executed on our exchanges.

Transaction-Based Expenses

Our equities and equity options markets pay fees to the SEC pursuant to Section 31 of the Exchange Act. Section 31 fees are recorded on a gross basis as a component of transaction and clearing fee revenue. These Section 31 fees are assessed to recover the government's costs of supervising and regulating the securities markets and professionals and are subject to change. We, in turn, collect corresponding activity assessment fees from member organizations clearing or settling trades on the equities and options exchanges, and recognize these amounts in our transaction and clearing revenues when invoiced. The activity assessment fees are designed to equal the Section 31 fees. As a result, activity assessment fees and the corresponding Section 31 fees do not have an impact on our net income, although the timing of payment by us will vary from collections. Section 31 fees were \$320 million and \$169 million for the six months ended June 30, 2020 and 2019, respectively, and \$154 million and \$100 million for the three months ended June 30, 2020 and 2019, respectively. The fees we collect are included in cash at the time of receipt and we remit the amounts to the SEC semi-annually as required. The total amount is included in accrued liabilities and was \$316 million as of June 30, 2020.

We make liquidity payments to cash and options trading customers, as well as routing charges made to other exchanges which are included in transaction-based expenses. We incur routing charges when we do not have the best bid or offer in the market for a security that a customer is trying to buy or sell on one of our securities exchanges. In that case, we route the customer's order to the external market center that displays the best bid or offer. The external market center charges us a fee per share (denominated in tenths of a cent per share) for routing to its system. We record routing charges on a gross basis as a component of transaction and clearing fee revenue. Cash liquidity payments, routing and clearing fees were \$807 million and \$480 million for the six months ended June 30, 2020 and 2019, respectively, and \$417 million and \$236 million for the three months ended June 30, 2020 and 2019, respectively.

Operating Expenses, Operating Income and Operating Margin

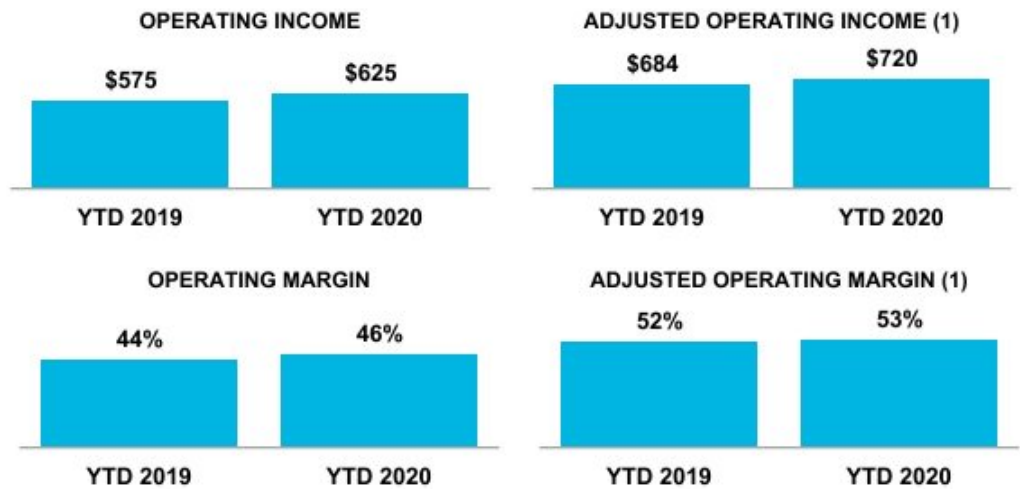
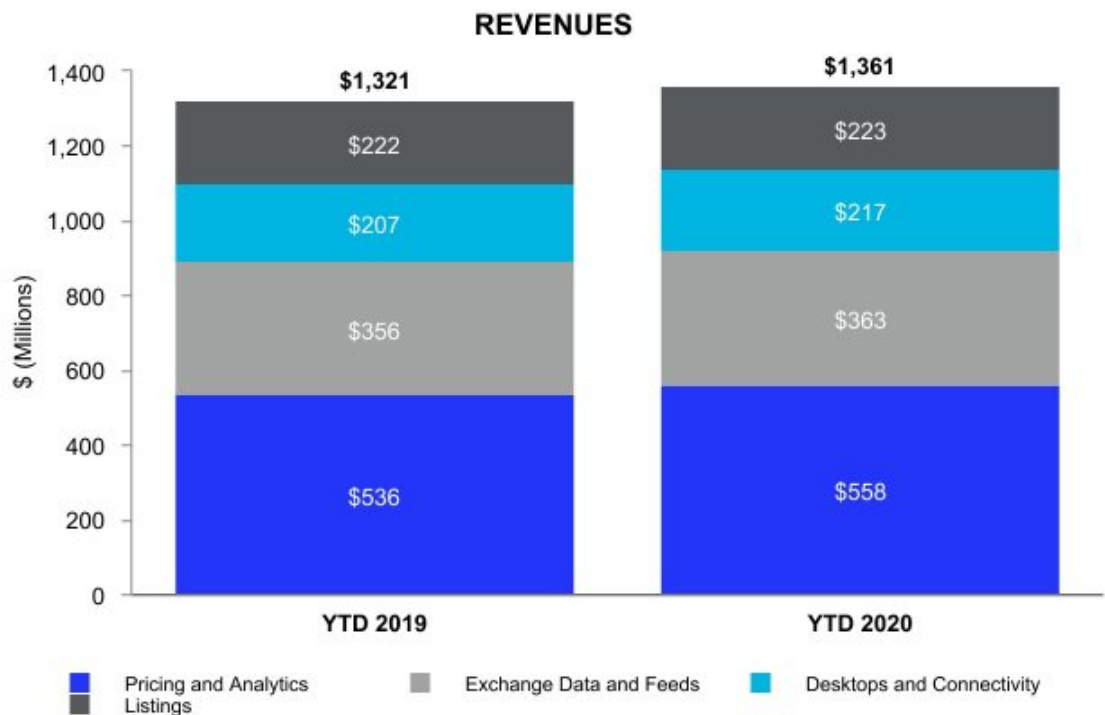
The following chart summarizes our Trading and Clearing segment's operating expenses, operating income and operating margin (dollars in millions). See "- Consolidated Operating Expenses" below for a discussion of the significant changes in our operating expenses.

Trading and Clearing Segment:	Six Months Ended June 30,			Change	Three Months Ended June 30,			Change
	2020	2019			2020	2019		
Operating expenses	\$ 592	\$ 477	24	%	\$ 282	\$ 249	13	%
Adjusted operating expenses ⁽¹⁾	\$ 531	\$ 431	23	%	\$ 256	\$ 226	14	%
Operating income	\$ 1,001	\$ 770	30	%	\$ 428	\$ 385	11	%
Adjusted operating income ⁽¹⁾	\$ 1,062	\$ 816	30	%	\$ 454	\$ 408	11	%
Operating margin	63 %	62 %	1 pt		60 %	61 %	(1 pt)	
Adjusted operating margin ⁽¹⁾	67 %	65 %	2 pts		64 %	64 %	—	

(1) The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

Data and Listings Segment

The following charts and table present our selected statements of income data for our Data and Listings segment (dollars in millions and YTD represents the six-month periods ended June 30th):



(1) The adjusted numbers in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See "- Non-GAAP Financial Measures" below.

	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Revenues:						
Pricing and analytics	\$ 558	\$ 536	4 %	\$ 282	\$ 270	4 %
Exchange data and feeds	363	356	2	183	180	2
Desktops and connectivity	217	207	5	109	103	6
Data services	1,138	1,099	4	574	553	4
Listings	223	222	—	111	111	—
Revenues	1,361	1,321	3	685	664	3
Other operating expenses	552	548	1	276	271	2
Depreciation and amortization	184	198	(8)	93	98	(6)
Operating expenses	736	746	(1)	369	369	—
Operating income	\$ 625	\$ 575	9 %	\$ 316	\$ 295	7 %

Our Data and Listings segment represents largely subscription-based, or recurring, revenues from data services and listings services offered across our trading and clearing businesses and ICE Data Services. Through ICE Data Services, we generate revenues from a range of global financial and commodity markets, including pricing and reference data, exchange data, analytics, feeds, index services, desktops and connectivity solutions. Through NYSE, NYSE American and NYSE Arca, we generate listings revenue related to the provision of listings services for public companies and ETFs, and related corporate actions for listed companies.

For both the six months and three months ended June 30, 2020 and the six months and three months ended June 30, 2019, 7% of our Data and Listings segment revenues were billed in pounds sterling or euros (all relating to our data services revenues). As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues denominated in foreign currencies changes accordingly. Due to the strengthening of the U.S. dollar compared to the pound sterling and euro, for the six months and three months ended June 30, 2020 as compared to the same periods in 2019, foreign currency fluctuations resulted in a decrease of \$2 million and \$1 million, respectively, to our Data and Listings segment revenues.

Our data services revenues are primarily subscription-based and increased 4% for both the six months and three months ended June 30, 2020 from the comparable periods in 2019. The increase in revenues was primarily due to the strong retention rate of existing customers, the addition of new customers, increased purchases by existing customers and increases in pricing of our products.

- **Pricing and Analytics:** Our pricing and analytics revenues increased 4% for both the six months and three months ended June 30, 2020 from the comparable periods in 2019. The increase in revenue was due to strength in our index businesses and continued growth in our pricing and reference data business driven by the strong retention rate of existing customers, the addition of new customers, increased purchases by existing customers and increases in pricing of our products. This growth was partially offset by \$1 million of unfavorable fluctuations in the U.S. dollar as compared to the pound sterling and euro for the six months and three months ended June 30, 2020 from the comparable period in 2019.
- **Exchange Data and Feeds:** Our exchange data and feeds revenues increased 2% for both the six months and three months ended June 30, 2020 from the comparable periods in 2019. The increase in revenue was largely due to continued growth in our futures exchange data and our consolidated feed offering, which was driven by the strong retention rate of existing customers, the addition of new customers and increases in pricing of our products.
- **Desktops and Connectivity:** Our desktop and connectivity revenues increased 5% and 6% for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019. The increase in revenue was driven primarily by growth in our connectivity services including the ICE Global Network, coupled with stronger desktop revenues.

Annual Subscription Value, or ASV, represents, at a point in time, the data services revenues subscribed for the succeeding 12 months. ASV does not include new sales, contract terminations or price changes that may occur during that 12-month period. ASV also does not include certain data services revenue streams that are not subscription-based. Revenue from ASV businesses has historically represented approximately 90% of our data revenues. Thus, while it is an indicative forward-looking metric, it does not provide a growth forecast of the next 12 months of data services revenues.

As of June 30, 2020, ASV was \$2.057 billion, which increased 4.2% compared to the ASV as of June 30, 2019. This does not adjust for year-over-year foreign exchange fluctuations or impacts of acquisitions.

Listings Revenues

Listings revenues in our securities markets arise from fees applicable to companies listed on our cash equities exchanges— original listing fees and annual listing fees. Original listing fees consist of two components: initial listing fees and fees related to corporate actions. Initial listing fees, subject to a minimum and maximum amount, are based on the number of shares that a company initially lists. All listings fees are billed upfront and the identified performance obligations are satisfied over time. Revenue related to the investor relations performance obligation is recognized ratably over the period these services are provided, with the remaining revenue recognized ratably over time as customers continue to list on our exchanges.

In addition, we earn corporate actions-related listing fees in connection with actions involving the issuance of new shares, such as stock splits, rights issues and sales of additional securities, as well as mergers and acquisitions. Listings fees related to other corporate actions are considered contract modifications of our listing contracts and are recognized ratably over time as customers continue to list on our exchanges.

Our listings revenues were flat for both the six months and three months ended June 30, 2020 compared to the comparable periods in 2019.

Operating Expenses, Operating Income and Operating Margin

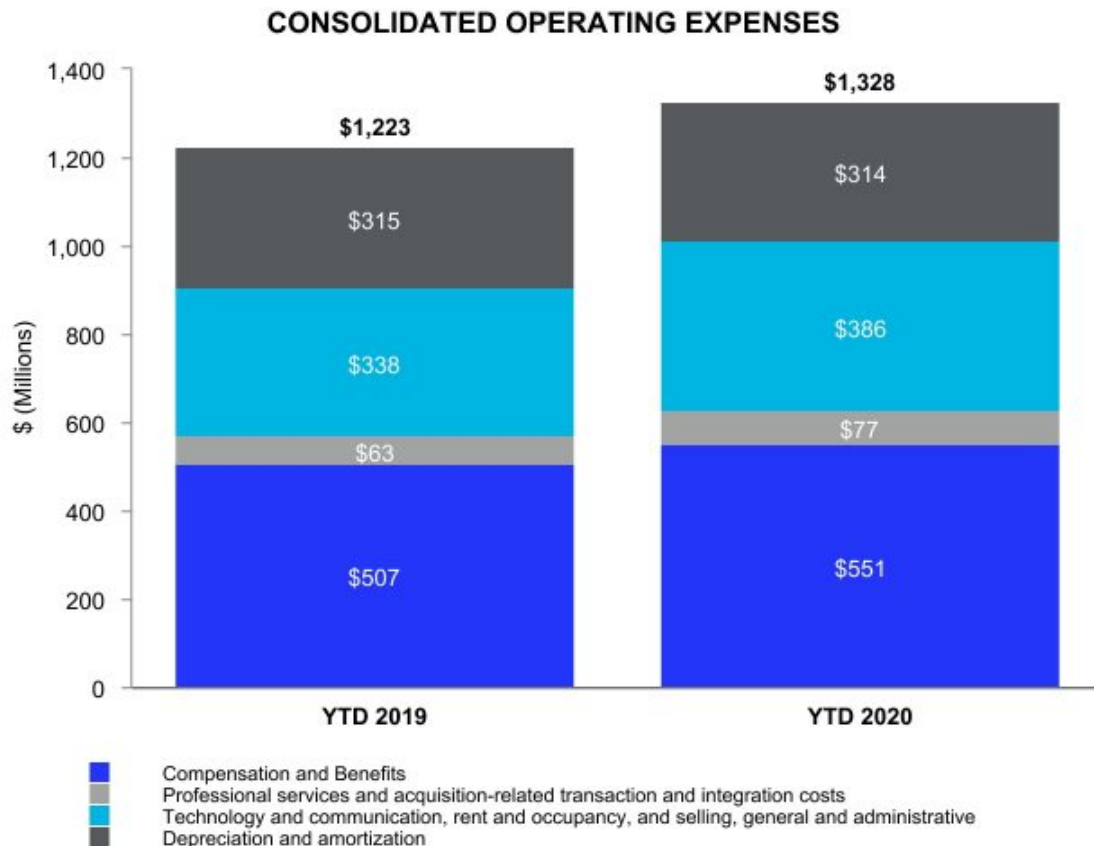
The following chart summarizes our Data and Listings segment's operating expenses, operating income and operating margin (dollars in millions). See “- Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses.

Data and Listings Segment:	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Operating expenses	\$ 736	\$ 746	(1) %	\$ 369	\$ 369	— %
Adjusted operating expenses ⁽¹⁾	\$ 641	\$ 637	1 %	\$ 319	\$ 314	1 %
Operating income	\$ 625	\$ 575	9 %	\$ 316	\$ 295	7 %
Adjusted operating income ⁽¹⁾	\$ 720	\$ 684	5 %	\$ 366	\$ 350	5 %
Operating margin	46 %	44 %	2 pts	46 %	44 %	2 pts
Adjusted operating margin ⁽¹⁾	53 %	52 %	1 pt	53 %	53 %	—

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

Consolidated Operating Expenses

The following presents our consolidated operating expenses (dollars in millions and YTD represents the six-month periods ended June 30th):



	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Compensation and benefits	\$ 551	\$ 507	9 %	\$ 273	\$ 259	5 %
Professional services	63	62	—	34	29	13
Acquisition-related transaction and integration costs	14	1	n/a	2	1	56
Technology and communication	257	220	17	126	113	12
Rent and occupancy	40	35	16	19	18	9
Selling, general and administrative	89	83	8	40	41	(2)
Depreciation and amortization	314	315	—	157	157	—
Total operating expenses	\$ 1,328	\$ 1,223	9 %	\$ 651	\$ 618	5 %

The majority of our operating expenses do not vary directly with changes in our volume and revenues, except for certain technology and communication expenses, including data acquisition costs, licensing and other fee-related arrangements and a portion of our compensation expense that is tied directly to our data sales or overall financial performance.

We expect our operating expenses to increase in absolute terms in future periods in connection with the growth of our business, and to vary from year-to-year based on the type and level of our acquisitions, our integrations and other investments.

For both the six months ended June 30, 2020 and 2019, 12% of our operating expenses were billed in pounds sterling or euros and for both the three months ended June 30, 2020 and 2019, 11% of our operating expenses were billed in pounds sterling or euros. Due to fluctuations in the U.S. dollar compared to the pound sterling and euro, our consolidated

operating expenses decreased \$4 million and \$2 million for the six months and three months ended June 30, 2020, respectively, from the comparable periods in 2019. See Item 3 “— Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” below for additional information.

Compensation and Benefits Expenses

Compensation and benefits expense is our most significant operating expense and includes non-capitalized employee wages, bonuses, non-cash or stock compensation, certain severance costs, benefits and employer taxes. The bonus component of our compensation and benefits expense is based on both our financial performance and individual employee performance. The performance-based restricted stock compensation expense is also based on our financial performance. Therefore, our compensation and benefits expense will vary year-to-year based on our financial performance and fluctuations in our number of employees. The below chart summarizes the significant drivers of our compensation and benefits expense results for the periods presented (dollars in millions, except employee headcount).

	Six Months Ended June 30,			Three Months Ended June 30,		
	2020	2019	Change	2020	2019	Change
Employee headcount	6,423	5,351	20 %			
Stock-based compensation expenses	\$ 61	\$ 64	(5) %	\$ 31	\$ 35	(13) %

Employee headcount increased for the six months ended June 30, 2020 from the comparable period in 2019 primarily due to 259 new employees at Bridge2 Solutions and 630 new employees in our ICE India office, who had previously been our contractors. Total compensation and benefits expenses increased for the six months and three months ended June 30, 2020 from the comparable periods in 2019 primarily due to \$26 million and \$15 million, respectively, in additional costs related to the acquisitions of Simplifile and Bridge2 Solutions and the establishment of ICE India, as well as increases of \$11 million and \$4 million, respectively, of expenses related to other increases in employee headcount and 2020 merit pay. The stock-based compensation expenses in the table above relate to employee stock option and restricted stock awards.

Professional Services Expenses

Professional services expense includes fees for consulting services received on strategic and technology initiatives, temporary labor, as well as regulatory, legal and accounting fees, and may fluctuate as a result of changes in the use of these services in our business.

Professional services expenses were flat for the six months ended June 30, 2020 from the comparable period in 2019. Professional services expenses increased for the three months ended June 30, 2020 from the comparable period in 2019 due to increased costs associated with regulatory and litigation matters.

Acquisition-Related Transaction and Integration Costs

We incurred \$14 million and \$2 million in acquisition-related transaction costs for the six months and three months ended June 30, 2020, respectively, primarily related to the Bakkt acquisition of Bridge2 Solutions. The Bridge2 Solutions acquisition costs include expenses of \$10 million resulting from a Bakkt incentive award market condition estimation adjustment that was directly related to the March 2020 capital call to fund the acquisition of Bridge2 Solutions. For both the six months and three months ended June 30, 2019, we incurred \$1 million in acquisition-related transaction and integration costs.

We expect to continue to explore and pursue various potential acquisitions and other strategic opportunities to strengthen our competitive position and support our growth. As a result, we may incur acquisition-related transaction costs in future periods.

Technology and Communication Expenses

Technology support services consist of costs for running our wholly-owned data centers, hosting costs paid to third-party data centers and maintenance of our computer hardware and software required to support our technology and cybersecurity. These costs are driven by system capacity, functionality and redundancy requirements. Communication expenses consist of costs for network connections for our electronic platforms and telecommunications costs.

Technology and communications expense also includes fees paid for access to external market data, licensing and other fee agreement expenses. Technology and communications expenses may be impacted by growth in electronic contract volume, our capacity requirements, changes in the number of telecommunications hubs and connections with customers to access our electronic platforms directly. Beginning in the second quarter of 2019, we have reflected amounts owed under certain third-party revenue share arrangements as technology and communication operating expenses rather than

as had been previously recorded net within transaction and clearing revenues, which resulted in an increase in technology and communications expense of \$14 million and \$3 million for the six months and three months ended June 30, 2020, respectively, as compared to the comparable period in 2019. Also, for the six months and three months ended June 30, 2020 total technology and communications expenses increased \$6 million and \$4 million, respectively, due to our acquisition of Simplifile in 2019 and Bridge2 Solutions and \$15 million and \$5 million, respectively, related to increased 2020 third-party revenue share fees.

Rent and Occupancy Expenses

Rent and occupancy expense relates to leased and owned property and includes rent, maintenance, real estate taxes, utilities and other related costs. We have significant operations located in and around Atlanta, New York, London and Hyderabad with smaller offices located throughout the world.

Rent and occupancy expenses increased for the six months ended June 30, 2020 from the comparable period in 2019 primarily due to the early termination of our NYSE Chicago office lease. Rent and occupancy expenses were flat for the three months ended June 30, 2020 from the comparable period in 2019.

Selling, General and Administrative Expenses

Selling, general and administrative expenses include marketing, advertising, public relations, insurance, bank service charges, dues and subscriptions, travel and entertainment, non-income taxes and other general and administrative costs.

Selling, general and administrative expenses increased for the six months ended June 30, 2020 from the comparable period in 2019 primarily due to a \$10 million charitable contribution in support of COVID-19 relief efforts, partially offset by lower travel expenses, marketing costs and non-income taxes. Selling, general and administrative expenses decreased for the three months ended June 30, 2020 from the comparable period in 2019 primarily due to lower travel expenses and marketing costs offset by accruals related to investigations and inquiries.

Depreciation and Amortization Expenses

Depreciation and amortization expense results from depreciation of long-lived assets such as buildings, leasehold improvements, aircraft, hardware and networking equipment, software, furniture, fixtures and equipment over their estimated useful lives. This expense includes amortization of intangible assets obtained in our acquisitions of businesses, as well as on various licensing agreements, over their estimated useful lives. Intangible assets subject to amortization consist primarily of customer relationships, trading products with finite lives and technology. This expense also includes amortization of internally-developed and purchased software over its estimated useful life.

We recorded amortization expenses on intangible assets acquired as part of our acquisitions, as well as on other intangible assets, of \$141 million and \$156 million for the six months ended June 30, 2020 and 2019, respectively, and \$71 million and \$79 million for the three months ended June 30, 2020 and 2019, respectively. Amortization expense decreased for the six months and three months ended June 30, 2020 from the comparable periods in 2019 as a result of certain Interactive Data intangible assets that became fully amortized in the fourth quarter of 2019.

We recorded depreciation expenses on our fixed assets of \$173 million and \$159 million for the six months ended June 30, 2020 and 2019, respectively, and \$86 million and \$78 million for the three months ended June 30, 2020 and 2019, respectively. Depreciation expense increased for the six months and three months ended June 30, 2020 from the comparable periods in 2019 primarily due to depreciation resulting from increased software development and networking equipment.

Consolidated Non-Operating Income (Expense)

Income and expenses incurred through activities outside of our core operations are considered non-operating. The following tables present our non-operating income (expenses) (dollars in millions):

	Six Months Ended June 30,				Three Months Ended June 30,		
	2020	2019	Change		2020	2019	Change
Other income (expense):							
Interest income	\$ 8	\$ 19	(56) %		\$ 2	\$ 10	(81) %
Interest expense	(156)	(142)	10		(84)	(71)	18
Other income, net	31	32	(5)		11	9	12
Total other income (expense), net	\$ (117)	\$ (91)	28 %		\$ (71)	\$ (52)	38 %
Net income attributable to non-controlling interest	\$ (13)	\$ (14)	(8) %		\$ (5)	\$ (6)	(17) %

Interest Income

Interest income decreased for the six months and three months ended June 30, 2020, from the comparable periods in 2019 primarily due to a decrease in short-term interest rates on various investments.

Interest Expense

Interest expense increased for the six months and three months ended June 30, 2020, from the comparable periods in 2019 primarily due to a \$14 million extinguishment payment we incurred related to the June 2020 early redemption of senior notes with an original maturity of December 1, 2020. See “- Debt” below.

Other income, net

We own a 40% interest in the OCC, which is regulated by the SEC and the CFTC. We recognized \$35 million and \$36 million during the six months ended June 30, 2020 and 2019, respectively, and \$18 million and \$9 million during the three months ended June 30, 2020 and 2019, respectively, of equity earnings as our share of the OCC's estimated profits, which is included in other income. During the six months ended June 30, 2019, equity earnings in the OCC included a \$19 million earnings adjustment in other income to reflect the disapproved OCC capital plan that had originally been established in 2015. Refer to Note 4 to our consolidated financial statements included in our 2019 Form 10-K for additional details on our OCC investment.

We incurred foreign currency transaction gains/(losses) of \$1 million and (\$1 million) for the six months ended June 30, 2020 and 2019, respectively, and (\$3 million) and \$3 million for the three months ended June 30, 2020 and 2019, respectively. This was primarily attributable to the fluctuations of the pound sterling and euro relative to the U.S. dollar. Foreign currency transaction gains and losses are recorded in other income, net, when the settlement of foreign currency assets, liabilities and payables that occur in non-functional currencies due to the increase or decrease in the period-end foreign currency exchange rates between periods. See Item 3 “- Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” included elsewhere in this Quarterly Report for more information on these items.

In 2016, the SEC approved a plan to establish a market-wide consolidated audit trail, or CAT, to improve regulators' ability to monitor trading activity with a phased implementation through 2019. Funding of the implementation and operation of the CAT was ultimately expected to be provided by both the execution venues, including us, and industry members. To date, however, funding has been provided solely by the execution venues in exchange for promissory notes expected to be repaid if industry member fees are: (i) approved by the SEC and (ii) collected to fund the CAT. Due to delays and failures in implementation and functionality by the original plan processor, as well as recently published proposals by the SEC for an amended timeline and implementation structure and the most recent CAT financial statements, we believe the risk that execution venues are not reimbursed has increased, resulting in a \$2 million CAT promissory note impairment during the three months ended June 30, 2020 related to the work performed by the original plan processor.

Non-controlling Interest

For consolidated subsidiaries in which our ownership is less than 100%, and for which we have control over the assets, liabilities and management of the entity, the outside stockholders' interests are shown as non-controlling interests. As of June 30, 2020, our non-controlling interests included those related to the non-ICE limited partners' interest in our CDS clearing subsidiaries, non-controlling interests in ICE Futures Abu Dhabi and redeemable non-controlling interests of the

non-ICE partners in Bakkt. During the three months ended June 30, 2020, we received a contribution from a group of minority investors for a non-controlling interest in ICE Futures Abu Dhabi.

In December 2018, Bakkt was capitalized with \$183 million in initial funding with ICE as the majority owner, along with a group of other minority investors, and in March 2020, an additional \$300 million in funding occurred with ICE maintaining its majority ownership. We hold a call option over these interests subject to certain terms. Similarly, the non-ICE partners in Bakkt hold a put option to require us to repurchase their interests subject to certain terms. These minority interests are reflected as redeemable non-controlling interests in temporary equity within our consolidated balance sheet and are subject to remeasurement when repurchase is probable.

Consolidated Income Tax Provision

Consolidated income tax expense was \$323 million and \$284 million for the six months ended June 30, 2020 and 2019, respectively, and \$145 million and \$150 million for the three months ended June 30, 2020 and 2019, respectively. The change in consolidated income tax expense between periods is primarily due to the tax impact of changes in our pre-tax income and the changes in our effective tax rate each period. Our effective tax rate was 21% and 23% for the six months ended June 30, 2020 and 2019, respectively, and 22% and 24% for the three months ended June 30, 2020 and 2019, respectively. The effective tax rates for the six months and three months ended June 30, 2020 are lower than the effective tax rates for the comparable periods in 2019 primarily due to enacted U.K. tax law that reduced the corporate tax rate from 19% to 17% effective April 1, 2020, as well as state apportionment changes. The U.K. government has subsequently enacted a reversal of this income tax reduction in July 2020 and we will account for the impact of that in the third quarter of 2020.

On March 27, 2020, the Coronavirus Aid, Relief, and Economic Security Act, or the CARES Act, was enacted and certain income tax related relief was provided under the CARES Act. There is no material impact of the CARES Act on our income tax provision for the six months and three months ended June 30, 2020.

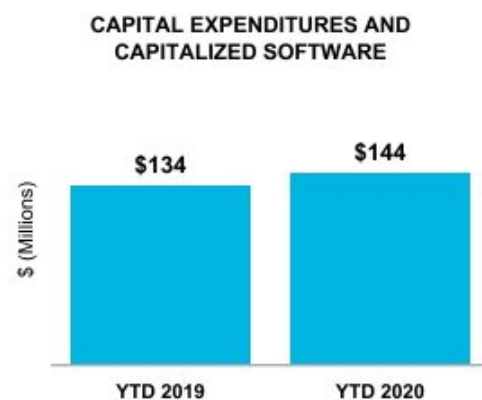
Quarterly Results of Operations

The following quarterly unaudited condensed consolidated statements of income data has been prepared on substantially the same basis as our audited consolidated financial statements and includes all adjustments, consisting only of normal recurring adjustments, necessary for the fair presentation of our consolidated results of operations for the quarters presented. The historical results for any quarter do not necessarily indicate the results expected for any future period. The following table sets forth quarterly consolidated statements of income data (in millions):

	Three Months Ended,				
	June 30, 2020	March 31, 2020	December 31, 2019	September 30, 2019	June 30, 2019
Revenues:					
Energy futures and options contracts	\$ 276	\$ 353	\$ 243	\$ 265	\$ 255
Agricultural and metals futures and options contracts	59	84	57	60	72
Financial futures and options contracts	76	123	80	91	78
Cash equities and equity options	672	669	442	401	410
Fixed income and credit	111	122	96	101	80
OTC and other transactions	13	13	11	11	12
Total transaction and clearing, net	1,207	1,364	929	929	907
Pricing and analytics	282	276	274	273	270
Exchange data and feeds	183	180	176	172	180
Desktops and connectivity	109	108	109	108	103
Total data services	574	564	559	553	553
Listings	111	112	113	114	111
Other revenues	74	75	66	67	63
Total revenues	1,966	2,115	1,667	1,663	1,634
Transaction-based expenses	571	556	369	327	336
Total revenues, less transaction-based expenses	1,395	1,559	1,298	1,336	1,298
Compensation and benefits	273	278	274	261	259
Professional services	34	29	28	35	29
Acquisition-related transaction and integration costs	2	12	1	—	1
Technology and communication	126	131	123	126	113
Rent and occupancy	19	21	16	17	18
Selling, general and administrative	40	49	45	33	41
Depreciation and amortization	157	157	189	158	157
Total operating expenses	651	677	676	630	618
Operating income	744	882	622	706	680
Other income (expense), net	(71)	(46)	(35)	(66)	(52)
Income tax expense	145	178	134	103	150
Net income	\$ 528	\$ 658	\$ 453	\$ 537	\$ 478
Net income attributable to non-controlling interest	(5)	(8)	(5)	(8)	(6)
Net income attributable to Intercontinental Exchange, Inc.	\$ 523	\$ 650	\$ 448	\$ 529	\$ 472

Liquidity and Capital Resources

Below are charts that reflect our capital allocation. The acquisition and integration costs in the chart below include cash paid for acquisitions, net of cash received for divestitures, cash paid for equity investments, cash paid for non-controlling interest and redeemable non-controlling interest, and acquisition-related transaction and integration costs, in each period (YTD represents the six-month periods ended June 30th).



We have financed our operations, growth and cash needs primarily through income from operations and borrowings under our various debt facilities. Our principal capital requirements have been to fund capital expenditures, working capital,

strategic acquisitions and investments, stock repurchases, dividends and the development of our technology platforms. We believe that our cash on hand and cash flows from operations will be sufficient to repay our outstanding debt, but we may also need to incur additional debt or issue additional equity securities in the future. See “- Future Capital Requirements” below.

For a discussion of the COVID-19 pandemic and how the pandemic may impact our business, results of operations or financial condition in the future, including our liquidity and capital resources, see “- Recent Developments”, above and “- Risk Factors” in Part 2, Item 1(A) below.

See “- Recent Developments” above for a discussion of the acquisition that we made during the six months ended June 30, 2020. This acquisition was funded from borrowing under our Commercial Paper Program along with cash flows from operations.

Our Commercial Paper Program enables us to borrow efficiently at reasonable short-term interest rates and provides us with the flexibility to de-lever using our strong annual cash flows from operating activities whenever our leverage becomes elevated as a result of investment or acquisition activities. We had net redemptions of \$611 million under our Commercial Paper Program during the six months ended June 30, 2020.

Upon maturity of our commercial paper and to the extent old issuances are not repaid by cash on hand, we are exposed to the rollover risk of not being able to issue new commercial paper. To mitigate this risk, we maintain an undrawn back-stop bank revolving credit facility, or the Credit Facility, for an aggregate amount which meets or exceeds the amount issued under our Commercial Paper Program at any time. If we were not able to issue new commercial paper, we have the option of drawing on the back-stop revolving facility. However, electing to do so would result in higher interest expense. For a discussion of our Commercial Paper Program and other indebtedness, see “- Debt” below.

Consolidated cash and cash equivalents were \$880 million and \$841 million as of June 30, 2020 and December 31, 2019, respectively. We had \$1.4 billion and \$1.3 billion in short-term and long-term restricted cash and cash equivalents as of June 30, 2020 and December 31, 2019, respectively.

As of June 30, 2020, the amount of unrestricted cash held by our non-U.S. subsidiaries was \$290 million. Due to U.S. tax reform, the majority of our foreign earnings since January 1, 2018 have been subject to immediate U.S. income taxation, and the existing non-U.S. unrestricted cash balance can be distributed to the U.S. in the future with no material additional income tax consequences.

Our cash and cash equivalents and financial investments are managed as a global treasury portfolio of non-speculative financial instruments that are readily convertible into cash, such as overnight deposits, term deposits, money market funds, mutual funds for treasury investments, short duration fixed income investments and other money market instruments, thus ensuring high liquidity of financial assets. We may invest a portion of our cash in excess of short-term operating needs in investment-grade marketable debt securities, including government or government-sponsored agencies and corporate debt securities. As of June 30, 2020, we held \$63 million of unrestricted cash that was set aside for legal, regulatory, and surveillance operations at NYSE.

In December 2019, our Board approved an aggregate of \$2.4 billion for future repurchases of our common stock with no fixed expiration date that became effective January 1, 2020. The \$2.4 billion replaced the previous amount approved by the Board. We expect this authorization to provide us with capacity for buybacks over six quarters and flexibility to act opportunistically. We expect funding for any share repurchases to come from our operating cash flow or borrowings under our Commercial Paper Program or our debt facilities.

We may discontinue stock repurchases at any time and may amend or terminate a Rule 10b5-1 trading plan at any time. The approval of our Board for the share repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board may increase or decrease the amount of capacity we have for repurchases from time to time.

Repurchases of our common stock may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. For the six months ended June 30, 2020, we repurchased 12.0 million shares of our outstanding common stock at a cost of \$1.1 billion, including 8.8 million shares at a cost of \$800 million under our Rule 10b5-1 trading plan and 3.2 million shares at a cost of \$299 million on the open market during an open trading period. For the six months ended June 30, 2019, we repurchased 10.0 million shares of our outstanding common stock at a cost of \$780 million, including 8.7 million shares at a cost of \$680 million under our Rule 10b5-1 trading plan and 1.3 million shares at a cost of \$100 million on the open market. The shares repurchased are held in treasury stock.

From time to time, we enter into Rule 10b5-1 trading plans, as authorized by our Board, to govern some or all of the repurchases of our shares of common stock. The timing and extent of future repurchases that are not made pursuant to a Rule 10b5-1 trading plan will be at our discretion and will depend upon many conditions. In making a determination

regarding any stock repurchases, management considers multiple factors, including overall stock market conditions, our common stock price performance, the remaining amount authorized for repurchases by our Board of Directors, the potential impact of a stock repurchase program on our corporate debt ratings, our expected free cash flow and working capital needs, our current and future planned strategic growth initiatives, and other potential uses of our cash and capital resources.

Cash Flow

The following table presents the major components of net changes in cash, cash equivalents, and restricted cash and cash equivalents (in millions):

	Six Months Ended June 30,	
	2020	2019
Net cash provided by (used in):		
Operating activities	\$ 1,378	\$ 1,382
Investing activities	(392)	(402)
Financing activities	(889)	(782)
Effect of exchange rate changes	(4)	1
Net increase in cash, cash equivalents and restricted cash and cash equivalents	\$ 93	\$ 199

Operating Activities

Net cash provided by operating activities primarily consists of net income adjusted for certain non-cash items, including depreciation and amortization and the effects of changes in working capital.

The \$4 million decrease in net cash provided by operating activities during the six months ended June 30, 2020 as compared to the prior period in 2019 was primarily a result of slower accounts receivable collections, primarily due to our decision to allow NYSE listed companies to delay payment of their annual invoices until later in the year, and higher income taxes paid primarily due to the U.K. government accelerating its required income tax installment payments in 2020 along with increased trading activity, offset by a \$216 million increase in net income. The remaining change is due to fluctuations in our working capital and the timing of various payments such as transaction-related expenses.

Investing Activities

Consolidated net cash used in investing activities for the six months ended June 30, 2020 relates to \$251 million in cash paid for acquisitions, net of cash acquired, \$48 million of capital expenditures and \$96 million of capitalized software development expenditures.

Consolidated net cash used in investing activities for the six months ended June 30, 2019 primarily relates to \$352 million cash paid for acquisitions (primarily Simplifile), \$57 million of capital expenditures, \$77 million of capitalized software development expenditures, a \$44 million return of capital related to our equity method investment in the OCC and proceeds from investments related to MERS. Refer to Note 4 to our consolidated financial statements included in this Quarterly Report for additional details on our OCC investment.

The capital expenditures primarily relate to hardware and software purchases to continue the development and expansion of our electronic platforms, data services and clearing houses, and leasehold improvements. The software development expenditures primarily relate to the continued development and expansion of our electronic trading platforms, data services and clearing houses.

Financing Activities

Consolidated net cash used in financing activities for the six months ended June 30, 2020 primarily relates to \$1.3 billion in repayments of our December 2020 Senior Notes, \$1.1 billion in repurchases of common stock, \$611 million in net redemptions under our Commercial Paper Program from a portion of the proceeds from the May 26, 2020 debt issuance, \$330 million in dividend payments to stockholders and \$72 million in cash payments related to treasury shares received for restricted stock tax payments and stock option exercises, partially offset by \$2.4 billion in net proceeds from our Senior Notes issued in May of 2020 (the 2030 Notes and the 2050 Notes).

Consolidated net cash used in financing activities for the six months ended June 30, 2019 primarily relates to \$780 million in repurchases of common stock, \$347 million in net borrowings under our Commercial Paper Program, \$312 million in dividend payments to our stockholders and \$60 million in cash payments related to treasury shares received for restricted stock tax payments and stock options exercises.

Debt

As of June 30, 2020, we had \$8.4 billion in outstanding debt, consisting of \$7.7 billion of senior notes, \$700 million under the Commercial Paper Program and \$5 million under a line of credit at our ICE India subsidiary. The commercial paper notes had original maturities ranging from six to 87 days as of June 30, 2020, with a weighted average interest rate of 0.37% per annum, and a weighted average remaining maturity of 42 days. As of December 31, 2019, we had \$7.8 billion in outstanding debt, consisting of \$6.5 billion of senior notes, \$1.3 billion under the Commercial Paper Program and \$10 million under a line of credit at our ICE India subsidiary. The commercial paper notes had original maturities ranging from two to 87 days as of December 31, 2019, with a weighted average interest rate of 1.84% per annum, and a weighted average remaining maturity of 22 days.

We currently have a \$3.4 billion senior unsecured revolving credit facility pursuant to a credit agreement with Wells Fargo Bank, N.A., as primary administrative agent, issuing lender and swing-line lender, Bank of America, N.A., as syndication agent, backup administrative agent and swing-line lender, and the lenders party thereto. As of June 30, 2020, of the \$3.4 billion that is currently available for borrowing under the Credit Facility, \$700 million is required to back-stop the amount outstanding under our Commercial Paper Program and \$171 million is required to support certain broker-dealer and other subsidiary commitments. The amount required to back-stop the amounts outstanding under the Commercial Paper Program will fluctuate as we increase or decrease our commercial paper borrowings. The remaining \$2.5 billion is available for working capital and general corporate purposes including, but not limited to, acting as a back-stop to future increases in the amounts outstanding under the Commercial Paper Program.

On May 26, 2020, we issued \$2.5 billion in aggregate principal amount of new senior notes, comprised of \$1.25 billion in aggregate principal amount of 2.10% senior notes due in 2030 and \$1.25 billion in aggregate principal amount of 3.00% Senior Notes due in 2050. We used the net proceeds from the offering for general corporate purposes, including to fund the redemption of our \$1.25 billion aggregate principal amount of 2.75% Senior Notes due in December 2020 and to pay down a portion of our commercial paper outstanding.

For additional details of our debt instruments, refer to Note 8 to our consolidated unaudited financial statements, included in this Quarterly report, and Note 10 to our consolidated financial statements included in our 2019 Form 10-K.

Future Capital Requirements

Our future capital requirements will depend on many factors, including the rate of growth across our Trading and Clearing and Data and Listings segments, strategic plans and acquisitions, available sources for financing activities, required and discretionary technology and clearing initiatives, regulatory requirements, the timing and introduction of new products and enhancements to existing products, the geographic mix of our business and potential stock repurchases.

We currently expect to incur capital expenditures (including operational and real estate capital expenditures) and to incur software development costs that are eligible for capitalization ranging in the aggregate between \$350 million and \$370 million in 2020, which we believe will support the enhancement of our technology, business integration and the continued growth of our businesses.

In December 2019, our Board approved an aggregate of \$2.4 billion for future repurchases of our common stock with no fixed expiration date that became effective on January 1, 2020. We expect this authorization to provide us with capacity for buybacks over six quarters and flexibility to act opportunistically. As of June 30, 2020, we had \$1.3 billion authorized for future repurchases of our common stock. Refer to Note 10 to our consolidated financial statements included in this Quarterly Report for additional details on our stock repurchase plan.

Our Board has adopted a quarterly dividend policy providing that dividends will be approved quarterly by the Board or the Audit Committee taking into account factors such as our evolving business model, prevailing business conditions, our current and future planned strategic growth initiatives and our financial results and capital requirements, without a predetermined net income payout ratio. During the second quarter of 2020, we paid a quarterly dividend of \$0.30 per share of our common stock for an aggregate payout of \$164 million, which includes the payment of dividend equivalents on unvested employee restricted stock units. On July 30, 2020, we announced a \$0.30 per share dividend for the third quarter of 2020 with the dividend payable on September 30, 2020 to stockholders of record as of September 16, 2020.

Other than the facilities for the ICE Clearing Houses, our Credit Facility and our Commercial Paper Program are currently the only significant agreements or arrangements that we have for liquidity and capital resources with third parties. See Notes 8 and 12 to our consolidated financial statements where discussed further. In the event of any strategic acquisitions, mergers or investments, or if we are required to raise capital for any reason or desire to return capital to our stockholders, we may incur additional debt, issue additional equity to raise necessary funds, repurchase additional shares

of our common stock or pay a dividend. However, we cannot provide assurance that such financing or transactions will be available or successful, or that the terms of such financing or transactions will be favorable to us. See “—Debt” above.

Non-GAAP Measures

We use certain financial measures internally to evaluate our performance and make financial and operational decisions that are presented in a manner that adjusts from their equivalent GAAP measures or that supplement the information provided by our GAAP measures. We use these adjusted results because we believe they more clearly highlight trends in our business that may not otherwise be apparent when relying solely on GAAP financial measures, since these measures eliminate from our results specific financial items that have less bearing on our core operating performance.

We use these measures in communicating certain aspects of our results and performance, including in this Quarterly Report, and believe that these measures, when viewed in conjunction with our GAAP results and the accompanying reconciliation, can provide investors with greater transparency and a greater understanding of factors affecting our financial condition and results of operations than GAAP measures alone. In addition, we believe the presentation of these measures is useful to investors for making period-to-period comparisons of results because the adjustments to GAAP are not reflective of our core business performance.

These financial measures are not presented in accordance with, or as an alternative to, GAAP financial measures and may be different from non-GAAP measures used by other companies. We encourage investors to review the GAAP financial measures included in this Quarterly Report, including our consolidated financial statements, to aid in their analysis and understanding of our performance and in making comparisons.

The table below outlines our adjusted operating expenses, adjusted operating income, adjusted operating margin, adjusted net income attributable to ICE common stockholders and adjusted earnings per share, which are non-GAAP measures that are calculated by making adjustments for items we view as not reflective of our cash operations and core business performance. These measures, including the adjustments and their related income tax effect and other tax adjustments (in millions, except for percentages and per share amounts), are as follows:

	Trading and Clearing Segment		Data and Listings Segment		Consolidated	
	Six Months Ended June 30,		Six Months Ended June 30,		Six Months Ended June 30,	
	2020	2019	2020	2019	2020	2019
Total revenues, less transaction-based expenses	\$ 1,593	\$ 1,247	\$ 1,361	\$ 1,321	\$ 2,954	\$ 2,568
Operating expenses	592	477	736	746	1,328	1,223
Less: Amortization of acquisition-related intangibles	51	46	90	109	141	155
Less: Acquisition-related success fee	10	—	—	—	10	—
Less: Accruals related to investigations and inquiries	—	—	5	—	5	—
Adjusted operating expenses	\$ 531	\$ 431	\$ 641	\$ 637	\$ 1,172	\$ 1,068
Operating income	\$ 1,001	\$ 770	\$ 625	\$ 575	\$ 1,626	\$ 1,345
Adjusted operating income	\$ 1,062	\$ 816	\$ 720	\$ 684	\$ 1,782	\$ 1,500
Operating margin	63 %	62 %	46 %	44 %	55 %	52 %
Adjusted operating margin	67 %	65 %	53 %	52 %	60 %	58 %
Net income attributable to ICE common stockholders					\$ 1,173	\$ 956
Add: Amortization of acquisition-related intangibles					141	155
Add: Extinguishment of 2020 Senior Notes					14	—
Add: Acquisition-related success fee					10	—
Add: Accruals related to investigations and inquiries					5	—
Add: Impairment of CAT promissory notes					2	—
Less: Income tax effect for the above items					(43)	(40)
Less: Deferred tax adjustments on acquisition-related intangibles					(10)	(13)
Add: Other tax adjustments					—	3
Adjusted net income attributable to ICE common stockholders					\$ 1,292	\$ 1,061
Basic earnings per share attributable to ICE common stockholders					\$ 2.14	\$ 1.69
Diluted earnings per share attributable to ICE common stockholders					\$ 2.13	\$ 1.68
Adjusted basic earnings per share attributable to ICE common stockholders					\$ 2.35	\$ 1.88
Adjusted diluted earnings per share attributable to ICE common stockholders					\$ 2.34	\$ 1.87

	Trading and Clearing Segment		Data and Listings Segment		Consolidated	
	Three Months Ended June 30,		Three Months Ended June 30,		Three Months Ended June 30,	
	2020	2019	2020	2019	2020	2019
Total revenues, less transaction-based expenses	\$ 710	\$ 634	\$ 685	\$ 664	\$ 1,395	\$ 1,298
Operating expenses	282	249	369	369	651	618
Less: Amortization of acquisition-related intangibles	26	23	45	55	71	78
Less: Accruals related to investigations and inquiries	—	—	5	—	5	—
Adjusted operating expenses	\$ 256	\$ 226	\$ 319	\$ 314	\$ 575	\$ 540
Operating income	\$ 428	\$ 385	\$ 316	\$ 295	\$ 744	\$ 680
Adjusted operating income	\$ 454	\$ 408	\$ 366	\$ 350	\$ 820	\$ 758
Operating margin	60 %	61 %	46 %	44 %	53 %	52 %
Adjusted operating margin	64 %	64 %	53 %	53 %	59 %	58 %
Net income attributable to ICE common stockholders					\$ 523	\$ 472
Add: Amortization of acquisition-related intangibles					71	78
Add: Extinguishment of 2020 Senior Notes					14	—
Add: Accruals related to investigations and inquiries					5	—
Add: Impairment of CAT promissory notes					2	—
Less: Income tax effect for the above items					(22)	(20)
Add/(Less): Deferred tax adjustments on acquisition-related intangibles					(9)	4
Adjusted net income attributable to ICE common stockholders					\$ 584	\$ 534
Basic earnings per share attributable to ICE common stockholders					\$ 0.96	\$ 0.84
Diluted earnings per share attributable to ICE common stockholders					\$ 0.95	\$ 0.84
Adjusted basic earnings per share attributable to ICE common stockholders					\$ 1.07	\$ 0.95
Adjusted diluted earnings per share attributable to ICE common stockholders					\$ 1.07	\$ 0.94

Amortization of acquisition-related intangibles are included in non-GAAP adjustments as excluding these non-cash expenses provides greater clarity regarding our financial strength and stability of cash operating results. We include an adjustment for Bridge2 Solutions acquisition costs resulting from a Bakkt incentive award market condition estimation adjustment as an acquisition-related success fee. This adjustment was directly related to the March 2020 capital call to fund the acquisition of Bridge2 Solutions. Although we include acquisition-related transaction costs as part of our core business expenses, this adjustment is appropriate since we exclude those costs that are directly related to the announcement, closing, financing or termination of a transaction.

The extinguishment payment on the 2020 Senior Notes is included as a non-GAAP adjustment as it relates to the June 2020 early redemption of senior notes with an original maturity of December 1, 2020 as a result of our new senior notes offering in May 2020. These costs include both a make-whole redemption payment and duplicative interest and are not considered part of our normal operations. We also include accruals for investigations and inquiries as a non-GAAP adjustment as we do not consider these to be reflective of our normal operations.

In September 2019, we included a non-GAAP adjustment for impairment charges on work performed by the original plan processor on the CAT. During the three months ended June 30, 2020, we included an additional CAT promissory note impairment charge on work performed by a subsequent plan processor. In 2016, the SEC approved a plan to establish a market-wide CAT to improve regulators' ability to monitor trading activity with a phased implementation through 2019. Funding of the implementation and operation of the CAT was ultimately expected to be provided by both the execution venues, including us, and industry members. To date, however, funding has been provided solely by the execution venues in exchange for promissory notes expected to be repaid if industry member fees are: (i) approved by the SEC and (ii) collected to fund the CAT. Due to delays and failures in implementation and functionality by the original plan processor, as well as recently published proposals by the SEC for an amended timeline and implementation structure and the most recent CAT financial statements, we believe the risk that execution venues are not reimbursed has increased, resulting in this impairment. This charge represents costs incurred with the original plan processor. Consistent with past practice, this is included as a non-GAAP adjustment as this is not considered a part of our core business operations.

The income tax effects relating to all non-GAAP adjustments above are included as non-GAAP adjustments. We also include non-GAAP adjustments for deferred tax adjustments on acquisition-related intangibles. The adjustment of (\$10 million) for the six months ending June 30, 2020 was due to U.S. state apportionment changes and the adjustment of (\$13

million) for the six months ending June 30, 2019 was due to U.S. state tax law and state apportionment changes. The \$3 million other tax adjustment for the six months ended June 30, 2019 is for additional audit settlement payments primarily related to pre-acquisition tax matters in conjunction with our acquisition of NYSE in 2013.

For additional information on these items, refer to our consolidated financial statements included in this Quarterly Report and “—Consolidated Operating Expenses”, above.

Contractual Obligations and Commercial Commitments

During the six months ended June 30, 2020, there were no significant changes to our contractual obligations and commercial commitments from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2019 Form 10-K.

Off-Balance Sheet Arrangements

As described in Note 12 to our consolidated financial statements, which are included elsewhere in this Quarterly Report, certain clearing house collateral is reported off-balance sheet. As described in Note 3 of our 2019 Form 10-K, Bakkt custodial assets are reported off-balance sheet. We do not have any relationships with unconsolidated entities or financial partnerships, often referred to as structured finance or special purpose entities.

New and Recently Adopted Accounting Pronouncements

Refer to Note 2 to our consolidated financial statements, which are included elsewhere in this Quarterly Report, for information on the new and recently adopted accounting pronouncements that are applicable to us.

Critical Accounting Policies

During the six months ended June 30, 2020, there were no significant changes to our critical accounting policies and estimates from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2019 Form 10-K.

ITEM 3. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

As a result of our operating and financing activities, we are exposed to market risks such as interest rate risk, foreign currency exchange rate risk and credit risk. We have implemented policies and procedures designed to measure, manage, monitor and report risk exposures, which are regularly reviewed by the appropriate management and supervisory bodies.

Interest Rate Risk

We have exposure to market risk for changes in interest rates relating to our cash and cash equivalents, short-term and long-term restricted cash and cash equivalents, short-term and long-term investments and indebtedness. As of June 30, 2020 and December 31, 2019, our cash and cash equivalents, short-term and long-term restricted cash and cash equivalents and short-term and long-term investments were \$2.3 billion and \$2.2 billion, respectively, of which \$238 million and \$282 million, respectively, were denominated in pounds sterling, euros or Canadian dollars, and the remaining amounts are denominated in U.S. dollars. We do not use our investment portfolio for trading or other speculative purposes. A hypothetical 50% decrease in short-term interest rates would decrease annual pre-tax earnings by \$8 million as of June 30, 2020, assuming no change in the amount or composition of our cash and cash equivalents and short-term and long-term restricted cash and cash equivalents.

As of June 30, 2020, we had \$8.4 billion in outstanding debt, of which \$7.7 billion relates to our senior notes, which bear interest at fixed interest rates. The remaining amount outstanding of \$700 million relates to our Commercial Paper Program, which bears interest at fluctuating rates and, therefore, subjects us to interest rate risk, and a \$5 million line of credit at our ICE India subsidiary. A hypothetical 100 basis point increase in short-term interest rates relating to the amounts outstanding under our Commercial Paper Program as of June 30, 2020 would decrease annual pre-tax earnings by \$7 million, assuming no change in the volume or composition of our outstanding indebtedness and no hedging activity. See Part I, Item 2 “Management’s Discussion and Analysis of Financial Condition and Results of Operations - Debt” included in this Quarterly Report.

The interest rates on our Commercial Paper Program are currently evaluated based upon current maturities and market conditions. The weighted average interest rate on our Commercial Paper Program decreased from 1.84% as of December 31, 2019 to 0.37% as of June 30, 2020. The decrease in the Commercial Paper Program weighted average interest rate was primarily due to the decision by the U.S. Federal Reserve to decrease the federal funds short-term interest rate by 150 basis points in March 2020 due to the impact of COVID-19 on financial markets, which impacted the liquidity and pricing volatility for all commercial paper issuances. The effective interest rate of commercial paper issuances will continue to fluctuate based on the movement in short-term interest rates along with shifts in supply and demand within the commercial paper market.

Foreign Currency Exchange Rate Risk

As an international business, we are subject to foreign currency exchange rate risk. We may experience gains or losses from foreign currency transactions in the future given that a significant part of our assets and liabilities are recorded in pounds sterling, Canadian dollars or euros, and a significant portion of our revenues and expenses are recorded in pounds sterling or euros. Certain assets, liabilities, revenues and expenses of foreign subsidiaries are denominated in the local functional currency of such subsidiaries. Our exposure to foreign denominated earnings for the six months and three months ended June 30, 2020 and June 30, 2019 is presented by primary foreign currency in the following table (dollars in millions, except exchange rates):

	Six Months Ended June 30, 2020		Three Months Ended June 30, 2020		Six Months Ended June 30, 2019		Three Months Ended June 30, 2019	
	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro
Average exchange rate to the U.S. dollar in the current year period	1.2603	1.1018	1.2413	1.1011	1.2936	1.1296	1.2851	1.1237
Average exchange rate to the U.S. dollar in the same period in the prior year	1.2936	1.1296	1.2851	1.1237	1.3762	1.2105	1.3606	1.1918
Average exchange rate increase (decrease)	(3)%	(2)%	(3)%	(2)%	(6)%	(7)%	(6)%	(6)%
Foreign denominated percentage of:								
Revenues, less transaction-based expenses	8 %	6 %	7 %	6 %	8 %	5 %	8 %	5 %
Operating expenses	10 %	2 %	9 %	2 %	10 %	2 %	9 %	2 %
Operating income	6 %	9 %	4 %	9 %	6 %	8 %	6 %	7 %
Impact of the currency fluctuations ⁽¹⁾ on:								
Revenues, less transaction-based expenses	\$ (5)	\$ (5)	\$ (3)	\$ (2)	\$ (13)	\$ (10)	\$ (6)	\$ (4)
Operating expenses	\$ (3)	\$ (1)	\$ (2)	\$ —	\$ (8)	\$ (2)	\$ (4)	\$ (1)
Operating income	\$ (2)	\$ (4)	\$ (1)	\$ (2)	\$ (5)	\$ (8)	\$ (2)	\$ (3)

(1) Represents the impact of currency fluctuation for the six months and three months ended June 30, 2020 and June 30, 2019 compared to the same periods in the prior year.

We have a significant part of our assets, liabilities, revenues and expenses recorded in pounds sterling or euros. For the six months and three months ended June 30, 2020, 14% and 12%, respectively, of our consolidated revenues, less transaction-based expenses, were denominated in pounds sterling or euros, and for the six months and three months ended June 30, 2020, 12% and 11%, respectively, of our consolidated operating expenses were denominated in pounds sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues and expenses denominated in foreign currencies changes accordingly.

Foreign currency transaction risk related to the settlement of foreign currency denominated assets, liabilities and payables occurs through our operations, which are received in or paid in pounds sterling, Canadian dollars, or euros, due to the increase or decrease in the foreign currency exchange rates between periods. We incurred foreign currency transaction gains/(losses) of \$1 million and (\$1 million) for the six months ended June 30, 2020 and 2019, respectively, and (\$3 million) and \$3 million for the three months ended June 30, 2020 and 2019, respectively. The foreign currency transaction gains/ (losses) were primarily attributable to the fluctuations of the pound sterling and euro relative to the U.S. dollar. A 10% adverse change in the underlying foreign currency exchange rates as of June 30, 2020, assuming no change in the composition of the foreign currency denominated assets, liabilities and payables and assuming no hedging activity, would result in a foreign currency transaction loss of \$9 million.

We entered into foreign currency hedging transactions during the six months and three months ended June 30, 2020 as economic hedges to help mitigate a portion of our foreign exchange risk exposure and may enter into additional hedging

transactions in the future to help mitigate our foreign exchange risk exposure. Although we may enter into additional hedging transactions in the future, these hedging arrangements may not be effective, particularly in the event of imprecise forecasts of the levels of our non-U.S. denominated assets and liabilities.

We have foreign currency translation risk equal to our net investment in our foreign subsidiaries. The financial statements of these subsidiaries are translated into U.S. dollars using a current rate of exchange, with gains or losses included in the cumulative translation adjustment account, a component of equity. Our exposure to the net investment in foreign currencies is presented by primary foreign currencies in the table below (in millions):

	As of June 30, 2020		
	Position in pounds sterling	Position in Canadian dollars	Position in euros
Assets	£ 795	C\$ 1,171	€ 153
of which goodwill represents	601	406	92
Liabilities	82	762	55
Net currency position	£ 713	C\$ 409	€ 98
Net currency position, in \$USD	\$ 884	\$ 301	\$ 110
Negative impact on consolidated equity of a 10% decrease in foreign currency exchange rates	\$ 88	\$ 30	\$ 11

Foreign currency translation adjustments are included as a component of accumulated other comprehensive income/(loss) within our balance sheet. See the table below for the portion of equity attributable to foreign currency translation adjustments as well as the activity for the six months and three months ended June 30, 2020 included within our statement of other comprehensive income. The impact of the foreign currency exchange rate differences in the table below were primarily driven by fluctuations of the pound sterling as compared to the U.S. dollar which were 1.2400, 1.2414 and 1.3260 as of June 30, 2020, March 31, 2020 and December 31, 2019, respectively, and by fluctuations of the euro as compared to the U.S. dollar which were 1.1234, 1.1035 and 1.1212 as of June 30, 2020, March 31, 2020 and December 31, 2019, respectively.

	Changes in Accumulated Other Comprehensive Income (Loss) from Foreign Currency Translation Adjustments (in millions)
Balance, as of December 31, 2019	\$ (177)
Net current period other comprehensive income (loss)	(78)
Balance, as of June 30, 2020	\$ (255)

	Changes in Accumulated Other Comprehensive Income (Loss) from Foreign Currency Translation Adjustments (in millions)
Balance, as of March 31, 2020	\$ (267)
Net current period other comprehensive income (loss)	12
Balance, as of June 30, 2020	\$ (255)

The future impact on our business relating to the U.K. leaving the EU and the corresponding regulatory changes are uncertain at this time, including future impacts on currency exchange rates.

Credit Risk

We are exposed to credit risk in our operations in the event of a counterparty default. We limit our exposure to credit risk by rigorously selecting the counterparties with which we make our investments, monitoring them on an ongoing basis and executing agreements to protect our interests.

Clearing House Cash Deposit Risks

The ICE Clearing Houses hold material amounts of clearing member cash and cash equivalent deposits which are held or invested primarily to provide security of capital while minimizing credit, market and liquidity risks. Refer to Note 12 to our

consolidated financial statements for more information on the ICE Clearing Houses' cash and cash equivalent deposits, which were \$94.0 billion as of June 30, 2020. While we seek to achieve a reasonable rate of return which may generate interest income for our clearing members, we are primarily concerned with preservation of capital and managing the risks associated with these deposits. As the ICE Clearing Houses may pass on interest revenues (minus costs) to the clearing members, this could include negative or reduced yield due to market conditions. For a summary of the risks associated with these deposits and how these risks are mitigated, see Part II, Item 7(A) "Quantitative and Qualitative Disclosures About Market Risk" in our 2019 Form 10-K.

ITEM 4. CONTROLS AND PROCEDURES

(a) *Evaluation of Disclosure Controls and Procedures.* As of the end of the period covered by this report, an evaluation was carried out by our management, with the participation of our Chief Executive Officer and Chief Financial Officer, of the effectiveness of our disclosure controls and procedures (as defined in Rule 13a-15(e) under the Exchange Act). Based upon that evaluation, our Chief Executive Officer and Chief Financial Officer concluded that these disclosure controls and procedures were effective as of the end of the period covered by this report.

(b) *Changes in Internal Controls over Financial Reporting.* There were no changes in our internal controls over financial reporting that occurred during our most recent fiscal quarter that have materially affected, or are reasonably likely to materially affect, our internal controls over financial reporting. As a result, no corrective actions were taken.

PART II. Other Information

ITEM 1. LEGAL PROCEEDINGS

We are subject to legal proceedings, claims and investigations that arise in the ordinary course of our business. We record estimated expenses and reserves for those matters in circumstances when a loss contingency is considered probable and the related amount is reasonably estimable. Any such accruals may be adjusted as circumstances change. Assessments of losses are inherently subjective and involve unpredictable factors. We do not believe that the resolution of these legal matters, including the matters described in this Quarterly Report, will have a material adverse effect on our consolidated financial condition, results of operations, or liquidity. It is possible, however, that future results of operations for any particular quarterly or annual period could be materially and adversely affected by any developments relating to the legal proceedings, claims and investigations. See Part I, Item 3 "Legal Proceedings" in our 2019 Form 10-K and Note 13 to the consolidated financial statements and related notes, which are included in this Quarterly Report.

ITEM 1(A). RISK FACTORS

The following represents a material change in our risk factors from those disclosed in Part 1, Item 1(A), "Risk Factors" in our 2019 Form 10-K.

The COVID-19 pandemic could have an adverse effect on our business, results of operations and financial condition.

The coronavirus (COVID-19) pandemic has created economic and financial disruptions globally and has led governmental authorities to take unprecedented measures to mitigate the spread of the disease, including travel bans, border closings, business closures, quarantines and shelter-in-place orders, and to take actions designed to stabilize markets and promote economic growth.

From an operational perspective, the spread of COVID-19 has resulted in, and could continue to result in, temporary closures of our office facilities and the office facilities of our customers and our third-party vendors. We have taken preventive measures and implemented contingency plans, and currently most of our employees are working remotely, with a limited number of employees working on-site at our facilities. However, we cannot assure you that such measures will adequately protect our business, and an extended period of remote work arrangements could introduce operational risks, including, but not limited to, cybersecurity risk, and could strain our technological resources and business continuity plans. If one or more of the third-party vendors to whom we outsource certain material activities claim that they cannot perform due to a force majeure or experience operational failures as a result of the COVID-19 pandemic, it could have a material adverse effect on our business, results of operations and financial condition. Further, although we maintain contingency plans for events such as pandemic outbreaks, the further spread of COVID-19 could impact the availability of our executive officers who are necessary to conduct our business.

The extent of the impact of the COVID-19 pandemic on our business, financial condition and results of operations will depend largely on future developments, including the duration and spread of the outbreak, its severity and the actions taken to contain the disease or treat its impact, all of which are highly uncertain and cannot be predicted at this time. A prolonged economic downturn could have an adverse effect on our revenues related to certain activities such as a decline in demand for certain data products and a decline in IPOs. The COVID-19 pandemic could also have an adverse impact on our customers' businesses, risk management needs and ability to trade, and the resulting impact on our business will depend on future developments, which are highly uncertain and cannot be predicted. Due to the speed with which the situation is developing, we are not able at this time to estimate the impact of the COVID-19 outbreak on our business, but the adverse impact on our business, results of operations and financial condition could be material.

While governmental organizations are engaging in efforts to combat the spread and severity of COVID-19, these measures may not be effective. Moreover, actions taken by U.S. or other governmental authorities that are intended to ameliorate the macroeconomic or other effects of COVID-19, or delays in the announcement or implementation of regulatory measures that had been pending prior to the COVID-19 pandemic, have resulted in and may in the future result in regulatory uncertainty and could in turn impact our business. At this time, we cannot predict how legal and regulatory responses to concerns about COVID-19 will impact our business.

The COVID-19 pandemic has resulted in, and may continue to result in, significant disruption of, and volatility in the global financial markets. Our parent company relies on payment of dividends from its subsidiaries, and certain regulators are considering instituting restrictions on certain entities' ability to pay dividends. Although we believe that our existing sources of liquidity are sufficient to meet our cash requirements for the foreseeable future, we cannot assure you that we will not experience liquidity constraints. If, as a result of the COVID-19 pandemic, conditions in the lending, capital and other financial markets are unfavorable, our regulated subsidiaries are unable to pay dividends to the parent company for an extended period of time or our access to capital and other sources of funding is cut off or negatively impacted, we could face liquidity issues and the availability and terms of future borrowings, renewals or refinancings could be adversely affected.

In addition to the other information set forth in this Quarterly Report, including the information in the "- Regulation" section of Item 2 - Management's Discussion and Analysis of Financial Condition and Results of Operations, you should carefully consider the factors discussed under "Risk Factors" and the regulation discussion under "Business - Regulation" in our 2019 Form 10-K. These risks could materially and adversely affect our business, financial condition and results of operations. The risks and uncertainties in our 2019 Form 10-K are not the only ones facing us. Additional

risks and uncertainties not presently known to us, or that we currently believe to be immaterial, may also adversely affect our business. In addition, to the extent that COVID-19 adversely affects our business, financial condition or results of operations, it may also heighten other risks described in Part 1, Item 1(A) under the caption “Risk Factors” in our 2019 Form 10-K.

ITEM 2. UNREGISTERED SALES OF EQUITY SECURITIES AND USE OF PROCEEDS

Stock Repurchases

The table below sets forth the information with respect to purchases made by or on behalf of ICE or any “affiliated purchaser” (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock during the three months ended June 30, 2020.

Period (2020)	Total number of shares purchased (in thousands)	Average price paid per share	Total number of shares purchased as part of publicly announced plans or programs (in thousands) (1)	Approximate dollar value of shares that may yet be purchased under the plans or programs (in millions) (1)
April 1 - April 30	1,587	\$86.58	9,230	\$1,563
May 1 - May 31	1,351	\$93.25	10,581	\$1,437
June 1 - June 30	1,446	\$94.49	12,027	\$1,301
Total	4,384	\$91.24	12,027	\$1,301

We have provided in the table below additional supplemental information with respect to purchases made by or on behalf of ICE or any “affiliated purchaser” (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock during each of the three months ended March 31, 2020 and June 30, 2020.

Three months ended	Total number of shares purchased (in thousands)	Average price paid per share	Total number of shares purchased as part of publicly announced plans or programs (in thousands) (1)	Approximate dollar value of shares that may yet be purchased under the plans or programs (in millions) (1)
March 31, 2020	7,643	\$91.50	7,643	\$1,701
June 30, 2020	4,384	\$91.24	12,027	\$1,301
Total	12,027	\$91.40	12,027	\$1,301

(1) Refer to the Notes to our consolidated financial statements and related notes, which are included in our Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 and elsewhere in this Quarterly Report, for details on our stock repurchase plans.

ITEM 3. DEFAULTS UPON SENIOR SECURITIES

None.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

ITEM 5. OTHER INFORMATION

Not applicable.

ITEM 6. EXHIBITS

Exhibit Number		Description of Document
4.1	—	Second Supplemental Indenture dated as of May 26, 2020 between Intercontinental Exchange, Inc., as issuer, and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.2 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on May 26, 2020, File No. 001-36198).
4.2	—	Form of 2.100% Senior Notes due 2030 (included in Exhibit 4.1 and incorporated by reference to Exhibit 4.3 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on May 26, 2020, File No. 001-36198).
4.3	—	Form of 3.000% Senior Notes due 2050 (included in Exhibit 4.1 and incorporated by reference to Exhibit 4.4 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on May 26, 2020, File No. 001-36198).
31.1	—	Rule 13a-14(a)/15d-14(a) Certification of Chief Executive Officer.
31.2	—	Rule 13a-14(a)/15d-14(a) Certification of Chief Financial Officer.
32.1	—	Section 1350 Certification of Chief Executive Officer.
32.2	—	Section 1350 Certification of Chief Financial Officer.
101	—	The following materials from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended June 30, 2020, formatted in Inline XBRL (Extensible Business Reporting Language): (i) the Consolidated Balance Sheets, (ii) the Consolidated Statements of Income, (iii) the Consolidated Statements of Comprehensive Income, (iv) the Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest (v) Consolidated Statements of Cash Flows and (vi) Notes to Consolidated Financial Statements, tagged as blocks of text.
104	—	The cover page from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended June 30, 2020 formatted in Inline XBRL.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Report to be signed on its behalf by the undersigned, thereunto duly authorized.

Intercontinental Exchange, Inc.
(Registrant)

Date: July 30, 2020

By: /s/ Scott A. Hill
Scott A. Hill
Chief Financial Officer
(Principal Financial Officer)

CERTIFICATIONS

I, Jeffrey C. Sprecher, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Intercontinental Exchange, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: July 30, 2020

/s/ Jeffrey C. Sprecher

Jeffrey C. Sprecher
Chairman of the Board and
Chief Executive Officer

CERTIFICATIONS

I, Scott A. Hill, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Intercontinental Exchange, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: July 30, 2020

/s/ Scott A. Hill

Scott A. Hill

Chief Financial Officer

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Quarterly Report of Intercontinental Exchange, Inc. (the “Company”) on Form 10-Q for the period ended June 30, 2020 as filed with the Securities and Exchange Commission on the date hereof (the “Report”), I, Jeffrey C. Sprecher, Chairman and Chief Executive Officer of the Company, certify, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that, to the best of my knowledge:

- (1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
- (2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: July 30, 2020

/s/ Jeffrey C. Sprecher

Jeffrey C. Sprecher
Chairman of the Board and
Chief Executive Officer

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Quarterly Report of Intercontinental Exchange, Inc. (the “Company”) on Form 10-Q for the period ended June 30, 2020 as filed with the Securities and Exchange Commission on the date hereof (the “Report”), I, Scott A. Hill, Chief Financial Officer of the Company, certify, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that, to the best of my knowledge:

- (1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
- (2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: July 30, 2020

/s/ Scott A. Hill

Scott A. Hill

Chief Financial Officer